1) Intro and Recap

1.1) Complex Numbers

Let $z = a + jb$, here is a really quick recap of useful formulas

$$
\begin{aligned}
&\text{Magnitude:} && |z| = \sqrt{a^2 + b^2} \\
&\text{Phase:} && \angle z = \arctan \frac{b}{a} \\
&\text{Complex Conjugate:} && z^* = a - jb \\
&\text{Euler's Identity:} && e^{jw} = \cos(w) + j\sin(w) \\
&\text{Polar Representation:} && \cos(w) = \frac{e^{jw} + e^{-jw}}{2} \\
& && \sin(w) = \frac{e^{jw} - e^{-jw}}{2j}
\end{aligned}
$$

Moreover, let $z_i = a_i + jb_i$, then

$$
\begin{aligned}
z_1 + z_2 &= (a_1 + a_2) + j(b_1 + b_2) \\
z_1 z_2 &= (a_1 a_2 - b_1 b_2) + j(a_1 b_2 + a_2 b_1) \\
|z_1 z_2| &= |z_1||z_2| \\
\angle(z_1 z_2) &= \angle(z_1)\angle(z_2)
\end{aligned}
$$

We will often need to find the n roots of $z^n = x$, $z, x, n \in \mathbb{C}$. Here are the steps of the solution

1. Write x in polar form: $x = Ae^{j\varphi}$ with $A = |x|$ and $\varphi = \angle x$
2. Then the k roots are:

$$
z_k = A^{1/n} e^{j\dfrac{\varphi + 2k\pi}{n}}, \quad k = 0, \dots, n-1
$$

3. This means that the zeroes are on a circumference of radius $A^{1/n}$ and are spaced by $\frac{\varphi + 2k\pi}{n}$. To find the angles you just suppose to start from φ/n and from there you divide the circle in n equally spaced points.

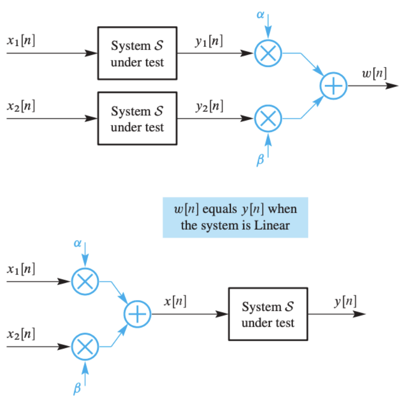
Since we often just need the unit circle, if $x \neq 1$ then this computation can be skipped.

Moreover recall that

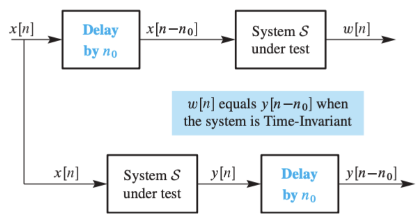
$$
\begin{aligned}
|a + b^{jk\omega}|^2 =& (a + be^{jk\omega})(a + be^{-jk\omega}) = a^2 + b^2 + 2ab\cos(k\omega) \\
& (a + be^{jk\omega})(a - be^{-jk\omega}) = a^2 + b^2 + j2ab\sin(k\omega)
\end{aligned}
$$

1.2) Signals And Systems

Definition 1 (Linearity Diagram)**.**



Definition 2 (Time-Invariance Diagram)**.**



Definition

Definition

1.3) Useful Results

DFT, DTFT, Frequency Response

$$\begin{aligned}
x[n] &= Ae^{j\phi}e^{j\hat{\omega}n} && \rightarrow\ y[n] = A|\mathcal{H}(\hat{\omega})|e^{j(\phi+\angle\mathcal{H}(\hat{\omega}))}e^{j\hat{\omega}n} \\
x[n] &= A\cos(\omega n+\phi) && \rightarrow\ y[n] = A|\mathcal{H}(\hat{\omega})|\cos(\omega n+\phi+\angle\mathcal{H}(\hat{\omega})) \\
x[n] &= A\sin(\omega n+\phi) && \rightarrow\ y[n] = A|\mathcal{H}(\hat{\omega})|\sin(\omega n+\phi+\angle\mathcal{H}(\hat{\omega}))
\end{aligned}$$

Z-Transform

$$H(z) = \frac{1}{1-az^{-}1} = -\frac{z}{a}\frac{a}{a-z}$$

And also

$$\begin{aligned}
H(z) &= \frac{1}{1-az^{-}1} \rightarrow h[n] = (a)^n u[n] && \text{ROC: } |z| > |a| \\
H(z) &= \frac{a}{a-z} \rightarrow h[n] = (a)^n u[n] && \text{ROC: } |z| < |a| \\
\text{that is: } H(z) &= \frac{z}{z-a} = -\frac{z}{a}\frac{a}{z-a} \rightarrow h[n] = -(a)^n u[-n-1] && \text{ROC: } |z| < |a|
\end{aligned}$$

Sums

$$\begin{aligned}
\sum_{k=0}^{\infty} p^k &= \frac{1}{1-p} \\
\sum_{k=0}^{\infty} k\cdot p^k &= \frac{p}{(1-p)^2} \\
1+z^{-1}+\ldots+z^{-N} = \sum_{k=0}^{N} z^{-k} &= \frac{1-z^{-(N+1)}}{1-z^{-1}}
\end{aligned}$$

2) Finite Impulse Response (FIR) Filters

As the name suggests, FIR filters have impulse response that settles to 0 in finite time. The general formula is the following:

$$\boxed{y[n] = \sum_{k=0}^{M} b_k x[n-k]}$$

Notice that b_k fully determines the filter so we can define the impulse response as $h[k] = \sum_{k=0}^{M} b_k \delta[n-k]$. This same relation is obtained by setting $x[n] = \delta[n]$:

$$y[n] = (\delta * h)[n] = h[n] = \sum_{k=0}^{M} b_k \delta[n-k]$$

We call M the **filter order** and $M+1$ the **filter length** (the amount of samples it uses). In general:

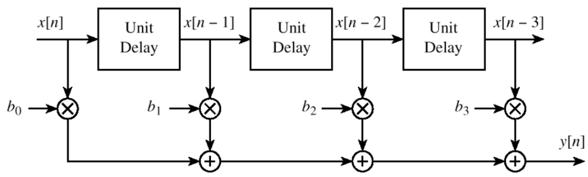
$$\begin{aligned} &\text{order} = \text{ max index } - \text{ min index} \\ &\text{length } = \text{ max index } - \text{ min index } + 1 = \text{ order } + 1 \end{aligned}$$

Convolution Sum

We have just seen the $*$ operator, this is called **convolution**.
An interpretation is the following:

- Time reverse $h[k]$
- Shift $h[-k]$ by n
- Form the product $v[k] = x[k]h[n-k]$
- Sum all samples of $v[k]$ to develop the n-th sample of $y[n]$

From here the general FIR structure



Diagram

Theorem 3 (Convolution Properties).

Commutative

$$(x * h)[n] = (h * x)[n]$$

Associative

$$(x * h)[n] * y[n] = x[n] * (y * h)[n]$$

Distributive

$$(x * (h + y))[n] = (x * h)[n] + (x * y)[n]$$

Dirac Delta Identity

$$(x * \delta)[n] = x[n]$$

Suppose to have a signal $x[n]$ with support N and a FIR with $h[n]$ that has order M, that is

$$\begin{aligned} x[n] &\neq 0 \quad 0 \leq n \leq N-1 \\ h[n] &\neq 0 \quad 0 \leq n \leq M \end{aligned}$$

Then the convolution will have:

$$\begin{aligned} \text{Support: } & 0 \leq n \leq (N-1)+(M) = N+M-1 \\ \text{Length: } & (N-1)+(M)+1 = N+M \end{aligned}$$

With a **group delay** of $\lfloor M/2 \rfloor$ samples

More in general, with lengths N, M we have

$$\begin{aligned} \text{Support: } & 0 \leq n \leq N+M-2 \\ \text{Length: } & N+M-1 \end{aligned}$$

Once the support is found the convolution can be easily calculated in each point since **the sum of indices of each sample product inside the convolution sum is equal to the index of the sample being generated by the convolution operation** by starting with $x[0]$:

- For example at index 3 we will have: $y[3] = x[0]h[3] + x[1]h[2] + x[2]h[1] + x[3]h[0]$
- At index i we have: $y[i] = \sum_{k=0}^{i} x[k]h[i-k]$

2.2) LTI FIR Sinusoidal Response

A LTI system can also be expressed in convolution form

$$y[n] = (x * h)[n]$$

In case of sinusoidal signals we can still obtain useful results, in fact we can define the DTFT:

> **Definition 4** (Frequency Response)**.**
>
> $$\mathcal{H}(\hat{w}) = \sum_{k=0}^{M} b_k e^{-j\hat{\omega}k} = \sum_{k=0}^{M} h[k] e^{-j\hat{\omega}k}$$
>
> This is a particular case of the Discrite Time Fourier Transform (DTFT) as it is the frequency-domain representation of the signal that was obtained with the impulse response of the system.

And get these formulas:

$$\begin{aligned} x[n] &= Ae^{j\phi}e^{j\hat{\omega}n} & &\rightarrow \ y[n] = A|\mathcal{H}(\hat{\omega})|e^{j(\phi+\angle\mathcal{H}(\hat{\omega}))}e^{j\hat{\omega}n} \\ x[n] &= A\cos(\omega n+\phi) & &\rightarrow \ y[n] = A|\mathcal{H}(\hat{\omega})|\cos(\omega n+\phi+\angle\mathcal{H}(\hat{\omega})) \\ x[n] &= A\sin(\omega n+\phi) & &\rightarrow \ y[n] = A|\mathcal{H}(\hat{\omega})|\sin(\omega n+\phi+\angle\mathcal{H}(\hat{\omega})) \end{aligned}$$

Proof:
Suppose to sample a sinusoidal from the continuous signal

$$x[n] = Ae^{j\phi}e^{j\hat{\omega}n} \overset{\text{sample}}{\leftarrow} x(t) = Ae^{j\phi}e^{j\omega n}$$

where $\hat{w} = \omega T_s$ with T_s the sampling period.
By applying the definition of FIR we have:

$$\begin{aligned} y[n] &= \sum_{k=0}^{M} h[k]x[n-k] \\ &= \sum_{k=0}^{M} h[k]e^{-j\hat{\omega}k}Ae^{j\phi}e^{j\hat{\omega}n} \\ &= \mathcal{H}(\hat{w})Ae^{j\phi}e^{j\hat{\omega}n} \end{aligned}$$

Since the frequency response is complex valued we can write it as $\mathcal{H}(\hat{\omega}) = |\mathcal{H}(\hat{\omega})|e^{j\angle\mathcal{H}(\hat{\omega})}$
And the sine/cos formulas are just a superposition and algebraic manipulation of this results by transforming the sinusoidal using Euler's formulas.

□

Also remember that:

$$w[n] = A\delta[n-k] \to \mathcal{W}(\hat{\omega}) = Ae^{j\hat{\omega}(-k)}$$

What if the input is suddenly applied with a units step?
Suppose the filter is of order M, then we can divide the signal in 3 parts: 0 if $n < 0$, the **transient** part with the first M samples and the **steady state part** with $n > M$

$$y[n] = \begin{cases} 0, & n < 0 \\ \sum_{k=0}^{n} (h[k]e^{-jwk})Ae^{jwn}, & 0 \le n < M \\ \sum_{k=0}^{M} (h[k]e^{-jwk})Ae^{jwn}, & n \ge M \end{cases}$$

Are there any properties of the **transient component** $y_t[n]$?

- FIR: $y_t[n] = 0$ outside of bounds
- IIR: $y_t[n] \overset{n\to\infty}{\to} 0$ if BIBO stable

Properties of the DTFT

Periodicity

$$\mathcal{H}(\hat{\omega} + 2\pi) = \mathcal{H}(\hat{\omega})$$

Proof

$$\mathcal{H}(\hat{\omega} + 2\pi) = \sum_{k=0}^{M} b_k e^{-j(\hat{\omega}+2\pi)k} = \sum_{k=0}^{M} b_k e^{-j\hat{\omega}k} = \mathcal{H}(\hat{\omega})$$

□

Conjugate Symmetry

$$\mathcal{H}(\hat{\omega}) = \mathcal{H}^*(-\hat{\omega})$$

Which is true for real coefficients: $b_k = b_k^*$.
This implies

$$|\mathcal{H}(-\hat{\omega})| = |\mathcal{H}(\hat{\omega})|$$
$$\angle\mathcal{H}(-\hat{\omega}) = -\angle\mathcal{H}(\hat{\omega})$$

$$\Re\{\mathcal{H}(\hat{\omega})\} = \Re\{\mathcal{H}(-\hat{\omega})\}$$
$$\Im\{\mathcal{H}(\hat{\omega})\} = -\{\mathcal{H}(-\hat{\omega})\}$$

Proof

$$\mathcal{H}(\hat{\omega}) = \sum_{k=0}^{M} b_k^* e^{j\hat{\omega}k} = \sum_{k=0}^{M} b_k e^{-j(-\hat{\omega})k} = \mathcal{H}(-\hat{\omega})$$

□

Cascaded LTI Systems

$$h[n] = (h_1 * h_2)[n] \xrightarrow{\mathcal{Z}} \mathcal{H}(\hat{\omega}) = \mathcal{H}_1(\hat{\omega})\mathcal{H}_2(\hat{\omega})$$

2.3) Examples of FIR Filters

Delay Filter

Theorem 5 (Delay Filter)**.**

This filter shifts the signal by an amount n_0

$$y[n] = x[n - n_0]$$
$$h[n] = \delta[n - n_0] \rightarrow \mathcal{H}(\hat{\omega}) = e^{-jn_0\hat{\omega}}$$

A delay of k will have the first k coefficients equal to 0 and the last equal to 1

$$b_k = \{0_0, 0_1, \ldots, 0_{k-1}, 1_k\}$$

L-Point Running Average

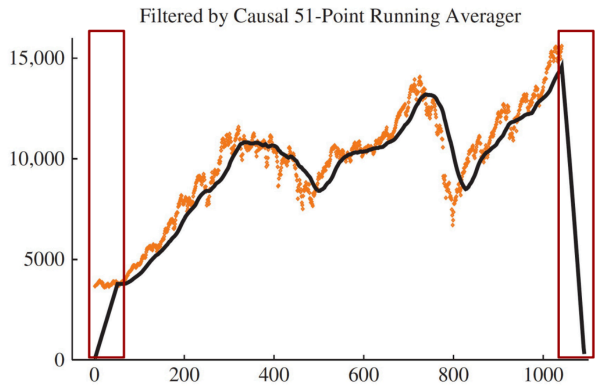
Definition 6 (Running Average Filter)**.**

An n-point running average filter is a filter that returns the mean of n values of the input, it can be:

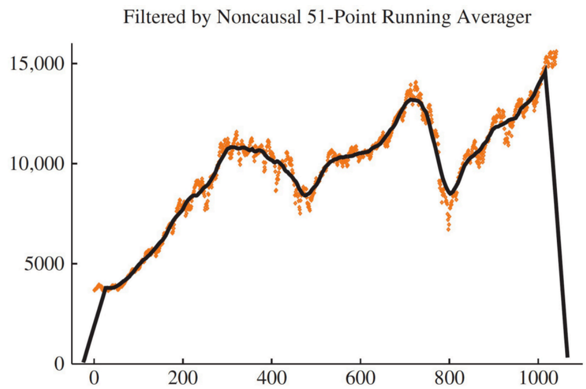
- Forwards:
$$y[n] = \frac{1}{l+1}\sum_{i=0}^{l} x[n+i]$$
This is a **anti-causal** filter and causes a shift of $M/2$ samples forwards
- Backwards:
$$y[n] = \frac{1}{l+1}\sum_{i=0}^{l} x[n-i]$$
This is a **causal** filter and causes a shift of $M/2$ samples backwards
- centralized (l odd):
$$y[n] = \frac{1}{l+1}\sum_{i=-l/2}^{l/2} x[n-i]$$

Now I will do some practical examples

Time Shift Example And Solution
Suppose to have a Backwards Running Avg Filter with $L = 51$, that is $y[n] = \frac{1}{51}\sum_{0}^{50} x[n-k]$
The first image shows the shift of $L/2$ samples. We use this filter since it is causal, moreover if we want to plot the output without the shift we can use the centralized version of the filter: $\tilde{y}[n] = \frac{1}{51}\sum_{-25}^{25} x[n-k]$ this gives the output of the second image.



Causal FIR Filter



Non Causal Filter

We can notice that the second filter is a cascade of the backwards filter with a dirac delta system.

$$\tilde{y}[n] = y[n] * \delta[n+25] = (x*h)[n] * \delta[n+25] = x[n] * (h[n] * \delta[n+25]) = x[n] * h[n+25]$$

And thus the impulse response of the cascade is $h[n+25]$.

2D Signals Filtering
In general 2D filters is a function with two variables, that is one for each dimension:

$$y[m,n] = \sum_{k=-M}^{M} \sum_{l=-N}^{N} b_{k,l} x[k-M, n-l]$$

Recall that for 2D signals causality is rarely important.

- **Edge Detection**
  This filter highlights edges by associating them with a high value in output magnitude.

The case of a 1D edge detection is very simple:

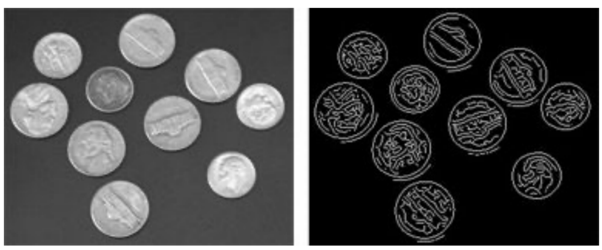
$$y[n] = x[n] - x[n-1]$$

here it is clear that similar values tend to cancel each other out and an increase in values comes with a positive increase.

Edge-finding filters are high pass filters since edges essentially highlight high frequency changes in the image. These filters can be represented by a matrix of the b_k values. This matrix has at the center the current sample $[m,n]$ and the other values are the corresponding surrounding samples $[m-i, n-j]$. A common matrix for edge detection is:

$$\begin{bmatrix} 1/4 & -1 & 1/4 \\ -1 & 3 & -1 \\ 1/4 & -1 & 1/4 \end{bmatrix}$$

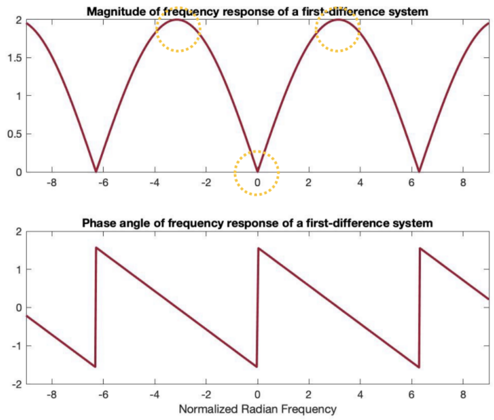
In both cases the average value of b_k is 0 in order to have similar values cancel out.


Example

The 1D system has frequency response

$$\mathcal{H}(\hat{\omega}) = 1 - e^{-j\hat{\omega}} = 1 - \cos\hat{\omega} + j\sin\hat{\omega} \rightarrow \begin{cases} |\mathcal{H}(\hat{\omega})| = 2|\sin\frac{\hat{\omega}}{2}| \\ \angle\mathcal{H}(\hat{\omega}) = \arctan\left(\frac{\sin\hat{\omega}}{1-\cos\hat{\omega}}\right) \end{cases}$$

By recalling the periodicity of the DTFT the following graphs show the High-Pass Filter behaviour



Graphs

- **Noise Removal**
  Let the L-point running avg filter be defined by

$$y[n] = \frac{1}{L}\sum_{k=0}^{L-1} x[n-k] \longrightarrow \mathcal{H}(\hat{\omega}) = \frac{1}{L}\sum_{k=0}^{L-1} e^{-j\hat{\omega}k}$$

By recalling that the sum of the first L elements is

$$\sum_{k=0}^{L-1} a^k = \frac{1-a^L}{1-a} \overset{a=e^{-j\hat{\omega}}}{\longrightarrow} \frac{1-e^{-j\hat{\omega}L}}{1-e^{-j\hat{\omega}}} = \frac{e^{-\frac{-j\hat{\omega}L}{2}}\sin\frac{\hat{\omega}L}{2}}{e^{-\frac{-j\hat{\omega}}{2}}\sin\frac{\hat{\omega}}{2}}$$

And thus

$$\mathcal{H}(\hat{\omega}) = \left(\frac{\sin\frac{\hat{\omega}L}{2}}{L\sin\frac{\hat{\omega}}{2}}\right)e^{-j\frac{\hat{\omega}(L-1)}{2}} = D_L(\hat{\omega})e^{-j\frac{\hat{\omega}(L-1)}{2}}$$

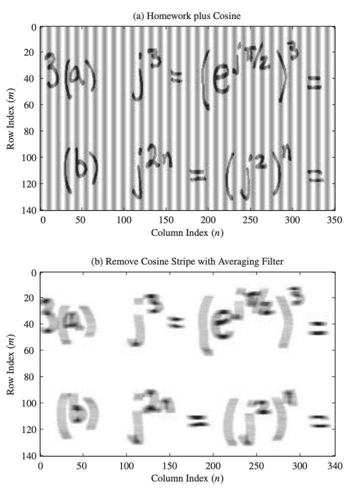
Where $D_L(\hat{\omega})$ is called **Dirichlet Form**

Suppose a 11 point running avg filter, then

$$\mathcal{H}(\hat{\omega}) = D_{11}(\hat{\omega})e^{-j\hat{\omega}5}$$

The filter will have a 0 at frequencies $\hat{\omega} = \dfrac{2\pi k}{11}$

It is possible to apply this filter to an image by doing every row individually. Let the image have k rows $x_k[n]$ and let there be a distortion $\tilde{x}_k[n] = x_k[n] + 100\cos\frac{2\pi n}{11}$, then the filter will blur the image but also completely remove the sinusoidal distortion since $|\mathcal{H}(2\pi/11)| = 0$.



Distortion Removal

Echo Removal

This is done by computing the **auto correlation**.
Define the correlation of two *real* sequences $x[n], y[n]$ as

$$R_{x,y}(n) = x[n] * y[-n] = x[-n] * y[n]$$

Then the autocorrelation just uses one signal:

$$R_x(n) = x[n] * x[-n]$$

Now suppose that the echo is obtained by doing having the original signal $w[n]$ and a delayed scaled version

$$x[n] = w[n] + aw[n - N]$$

By plotting the autocorrelation function we can determine the delay N as it will have peaks at $n = 0, n = N$ and $n = -N$.

3) Z-Transform

Definition 7 (Z-Transform)**.**

Bilateral:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n},\ z \in R_x \subset \mathbb{C}$$

Unilateral:

$$X_+(z) = \sum_{n=0}^{\infty} x[n]z^{-n},\ z \in R_{x^+} \subset \mathbb{C}$$

If the signal is causal these two forms are equivalent

R_x is called the Region Of Convergence (ROC), that is the values where the summation doesn't diverge, it holds that

$$R_x = \{z \in \mathbb{C} : R_- < |z| < R_+\}$$

and is thus circular. It can also be empty, the full $\mathbb{C}$ or an anular region.

We can distinguish 4 cases by the ROC-signal pair:

1. **x[n] with finite duration:** $0 < |z| < \infty$ that is the entire complex plane. If $b_0 < 0$ it doesn't converge at infinity. If $b_k > 0$ it doesn't converge in 0
2. **x[n] causal:** $\max_i |p_i| < |z|$ that is, the points outside of the circle of radius R_-
3. **x[n] anticausal:** $|z| < \min_i |p_i|$ that is, the points in the circle of radius R_+
4. **x[n] bilateral** $\max_j\{\text{causal } |p_j|\} < |z| < \min_i\{\text{anti-causal } |p_i|\}$ this is an anular region delimited by the two circles of radius R_- and R_+. If the causal/anti-causal poles don't satisfy the inequality we have empty ROC

By using D'Alembert it is possible to find R_- for causal signals

$$R_- = \lim_{n\to\infty} \left|\frac{x[n+1]}{x[n]}\right| < |z|$$

If $|z| = R_-$ no conclusion can be drawn and other methods must be used.

From the definition it is possible to obtain some results:

$$\begin{aligned}
x[n] &= \delta[n-k] & &\overset{X(z)}{\rightarrow} X(z) = z^{-k}, & R_x &= \begin{cases} \mathbb{C} & k = 0 \\ \mathbb{C}/\{0\} & k > 0 \\ \mathbb{C}/\{\infty\} & k > 0 \end{cases} \\
x[n] &= a^n u[n] & &\overset{X(z)}{\rightarrow} X(z) = \frac{1}{1-az^{-1}}, & R_x &= \{|a| < |z|\} \\
x[n] &= a^n u[-n] & &\overset{X(z)}{\rightarrow} X(z) = \frac{1}{1-a^{-1}z}, & R_x &= \{|z| < |a|\}
\end{aligned}$$

Where these results were obtained by recalling that

$$\sum_{k=0}^{\infty} p^k = \frac{1}{1-p}$$

Moreover if the z transform on the unit circle converges it corresponds to the Fourier transform:

$$X(z) \overset{z=e^{j\theta}}{=} X(e^{j\theta})$$

One important property of the z transform is that it can be written as a ratio of polynomials in order to better see the poles and zeroes of the transfer function

$$X(z) = \frac{P(z)}{Q(z)}$$

One important sum to remember is the following:

$$1 + z^{-1} + \ldots + z^{-N} = \sum_{k=0}^{N} z^{-k} = \frac{1 - z^{-(N+1)}}{1 - z^{-1}}$$

Properties of the z-Transform

In this chapter we will have $x[n], y[n], h[n]$ have z-Transforms $X(z), Y(z)$ with ROC R_X, R_Y, R_H with $R_X = \{z \in \mathbb{C} : R_- < |z| < R_+\}$ and $\alpha, \beta \in \mathbb{C}$ and $n_0 \in \mathbb{N}^+/\{0\}$

Linearity

$$w[n] = \alpha x[n] + \beta y[n] \xrightarrow{\mathcal{Z}} W(z) = \alpha X(z) + \beta Y(z) \qquad R_X \cap R_Y \subseteq R_W$$

Time-Delay

$$w[n] = x[n - n_0] \xrightarrow{\mathcal{Z}} W(z) = z^{-n_0} X(z) \qquad R_W = R_X$$

Proof:

$$W(z) = \sum_{n=-\infty}^{\infty} x[n - n_0] z^{-n} \overset{k=n-n_0}{=} \sum_{k=-\infty}^{\infty} x[n] z^{-n-n_0} = z^{-n_0} \sum_{k=-\infty}^{\infty} x[n] z^{-n} = z^{-n_0} X(z)$$

□

Bilateral Time Advance

$$w[n] = x[n + n_0] \xrightarrow{\mathcal{Z}} W(z) = z^{n_0} X(z) \qquad R_W = R_X$$

same proof as before.

Unilateral Time Advance

$$w[n] = x[n + n_0] \xrightarrow{\mathcal{Z}} W_+(z) = z^{n_0} X_+(z) - z^{n_0} \sum_{n=0}^{n_0 - 1} x[n] z^{-n} \qquad R_W = R_X$$

Proof:

$$\begin{aligned} W_+(z) &= \sum_{n=0}^{\infty} x[n - n_0] z^{-n} \overset{k=n-n_0}{=} \sum_{k=n_0}^{\infty} x[n] z^{-n-n_0} \\ &= z^{-n_0} \left(\sum_{k=0}^{\infty} x[n] z^{-n} + \sum_{k=0}^{n_0-1} x[n]^{-n} \right) = z^{-n_0} \left(X_+(z) + \sum_{k=0}^{n_0-1} x[n]^{-n} \right) \end{aligned}$$

□

Time Reversal

$$w[n] = x[-n] \xrightarrow{\mathcal{Z}} W(z) = X(z^{-1}) \qquad R_W = \left\{ z \in \mathbb{C} : \frac{1}{R_+} < |z| < \frac{1}{R_-} \right\}$$

Convolution

$$w[n] = (x * h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] = \sum_{k=-\infty}^{\infty} x[n-k]h[k] \xrightarrow{\mathcal{Z}} W(z) = X(z)H(z) \quad R_X \cap R_H \subseteq R_W$$

Bibo Stability

$$\text{BIBO stability} \iff \text{ROC includes unit circle, that is: } \{z \in \mathbb{C} : |z| = 1\} \subset R_H$$

Proof
A LTI system is BIBO stable if it's response $h[n]$ is absolutely summable, On the unit circle $z = e^{j\theta}$ we get:

$$\sum_{n=-\infty}^{\infty} |h[n]e^{-jn\theta}| = \sum_{n=-\infty}^{\infty} |h[n]| < +\infty$$

$\square$

Complex Conjugate

$$w[n] = x^*[n] \xrightarrow{\mathcal{Z}} W(z) = X^*(z^*) \qquad R_W = R_X$$

If the signal is real $x[n] = x^*[n]$ then the transform is hermitian $X(z) = X^*(z^*)$ and thus $W(z) = X(z)$

Differentiation

$$\frac{dX(z)}{dz} = -z^{-1} \sum_{n=-\infty}^{\infty} nx[n]z^{-n}$$

Moreover

$$\begin{aligned} w[n] = nx[n] &\xrightarrow{\mathcal{Z}} W(z) = -z\frac{dX(z)}{dz} && R_X = R_W \\ &\dots \\ w[n] = n^{n_0}x[n] &\xrightarrow{\mathcal{Z}} W(z) = -z^{n_0}\frac{d^{n_0}X(z)}{dz^{n_0}} && R_X = R_W \end{aligned}$$

Multiplication by Exponential Signal

$$w[n] = a^n x[n] \xrightarrow{\mathcal{Z}} W(z) = X\left(\frac{z}{a}\right) \qquad R_W = \{z \in \mathbb{C} : |a|R_- < |z| < |a|R_+\}$$

Theorem 8 (Initial Value Theorem)**.**

On the one sided z-transform we have that:

$$x[0] = \lim_{z \to +\infty} X_+(z)$$

Recall that for causal signals the one sided corresponds to the normal z-transform.
Proof:
$X_+(z) = x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$ *clearly the limit will make* $z^{-k} \to 0$

$\square$

Theorem 9 (Final Value Theorem)**.**

$$\lim_{x \to \infty} x[n] = \lim_{z \to 1} (z-1)X_+(z)$$

4) Linear Phase Filters

In many applications it is important to preserve the shape of the signal, that is:

$$y[n] = Ax[n - n_0]$$

From here we find that

$$H(z) = Az^{-n_0} \longleftrightarrow H(e^{j\theta}) = Ae^{-jn_0\theta}$$

Which clearly shows constant magnitude and linear phase:

$$|H(e^{j\theta}| = |A| \qquad \varphi(\theta) = \angle H(e^{j\theta}) = -n_0\theta + \begin{cases} 0 & \text{if } A > 0 \\ \pi & \text{if } A < 0 \end{cases}$$

And we introduce the **group delay** which is constant:

$$\tau_H(\theta) = -\frac{d\varphi_H(\theta)}{d\theta} = n_0$$

Remark.

These conditions just need to be held in the band of the signal

Theorem 10.

A discrete-time. LTI, real and causal system has linear phase frequency response if and only if its impulse response is symmetric (or antisymmetric), that is:

$$H(\theta) \text{ has linear phase} \iff h[n] = \pm h[N - n]$$

where only one of the $\pm$ is needed.

For now I will omit the proof

Corollary 11.

If a system has symmetric impulse response it's frequency response will be

$$H(e^{j\theta}) = \begin{cases} e^{-j\frac{N}{2}\theta}\overline{H}(\theta) & N \text{ even} \\ je^{-j\frac{N}{2}\theta}\overline{H}(\theta) & N \text{ odd} \end{cases} \qquad \varphi(\theta) = \begin{cases} -\frac{N}{2}\theta + \begin{cases} 0 \\ \pi \end{cases} & N \text{ even} \\ -\frac{N}{2}\theta + \frac{\pi}{2} + \begin{cases} 0 \\ \pi \end{cases} & N \text{ odd} \end{cases}$$

*With $\overline{H}(\theta)$ a even or odd real function called **amplitude response***

Notice how in both cases $\tau_H(\theta) = \frac{N}{2}$.

The order determines two possible scenarios:

- N even $\iff$ $h[n]$ odd: the center of symmetry is a sample of $h[n]$
- N odd $\iff$ $h[n]$ even: there is **no self symmetric sample** and the distortion condition is not satsified.

The no distortion condition has a **design constraint** as it halves the degrees of freedom available.

In general a LP filter can be written in the following form:

$$H(e^{j\theta}) = e^{-j\beta}2e^{-j\frac{N}{2}\theta}Q(\theta)P(\theta) \qquad P(\theta) = \sum_{k=0}^{r} p_k \cos(\theta n)$$

$$\beta = \begin{cases} 0 & h[n] \text{ symmetric} \\ \frac{\pi}{2} & h[n] \text{ antisymmetric} \end{cases} \qquad \begin{aligned} p_0 &= \begin{cases} h[\frac{N}{2}] & h[n] \text{ symmetric} \\ 0 & h[n] \text{ antisymmetric} \end{cases} \\ p_k &= \begin{cases} h[\frac{N}{2}-2] & h[n] \text{ symmetric} \\ h[\frac{N+1}{2}-2] & h[n] \text{ antisymmetric} \end{cases} \end{aligned}$$

	N	Symmetry	No Distortion	Q(θ)	Zeroes	Constraints	r
Type 1	even	symmetric	yes	1	$z_i \implies 1/z_i$		$N/2$
Type 2	odd	symmetric	no	$2\cos\frac{\theta}{2}$	$z_i \implies 1/z_i$, $z=-1$ is a zero	Only low pass	$(N-1)/2$
Type 3	even	antisymmetric	yes	$2\sin\theta$	$z_i \implies 1/z_i$, $z=\pm 1$ is a zero	No low pass	$(N-2)/2$
Type 4	odd	antisymmetric	no	$2\sin\frac{\theta}{2}$	$z_i \implies 1/z_i$, $z=1$ is a zero	No low pass	$(N-1)/2$

Remark.

A causal IIR system cannot satisfy these constraints (infinite and one sided thus can't be symmetric)

Here is the proof of the symmetry requirements
Define $\overline{H}(\theta)$ the magnitude part of $H(\theta)$, the we can rewrite the frequency response by forcing linear phase:

$$H(e^{j\omega}) = e^{j(\alpha\omega+\beta)}\overline{H}(\omega)$$

Recall that a real impulse response has frequency response with conjugate symmetry $H(e^{j\omega}) = H^*(e^{-j\omega})$ and thus we have even magnitude: $|H(e^{j\omega})| = |H(e^{-j\omega})|$. Therefore the magnitude $\overline{H}(\omega)$ is either a even or odd real function: $\overline{H}(\omega) = \pm\overline{H}(-\omega)$. Now we have

$$e^{j(\alpha\omega+\beta)}\overline{H}(\omega) = e^{j(\alpha\omega-\beta)}\overline{H}(-\omega)$$

- Case $\overline{H}$ even function:
  From the above function we obtain $e^{j\beta} = e^{-j\beta}$ and thus $\beta = 0, \pi$. Let's choose 0.
  Now notice the following:

$$\overline{H}(\omega) = \sum_{n=0}^{N} h[n]e^{-j\theta(\alpha+n))}$$
$$\overline{H}(-\omega) = \sum_{k=0}^{N} h[k]e^{-j\omega(\alpha+k)} \overset{k=N-n}{=} \sum_{n=0}^{N} h[N-n]e^{-j(\alpha+N-n)}$$

  and this shows clearly that $h[n] = h[N-n]$. Moreover we have $\alpha = N/2$.
- Case $\overline{H}$ odd function
  From the above function we obtain $e^{j\beta} = -e^{-j\beta}$ and thus $\beta = \pm\pi/2$.
  By the same steps as before we notice that it is the same just with the term $-j$ in front which forces the antisymmetry and the $\pi/2$ phase difference.

What about the zeroes?

- Case Type I and Type II (symmetric)
  Notice that due to time shift and time reversal we have

  $$H(z) = z^{-N} H(1/z)$$

  This is identical to it's mirror version. In particular $z_i = -1$ is important since

  $$H(-1) = (-1)^{-N} H(-1)$$

  With N even this is an identity, but with N odd we have that $H(-1) = -H(-1)$ there must be a 0. This means that Type I has $N/2$ degrees of freedom, while Type II has $(N-1)/2$.
- Case Type III and Type IV
  Notice that due to time shift and time reversal we have

  $$H(z) = -z^{-N} H(1/z)$$

  Now with $z_i = 1$ we have a 0 in any case, and for Type 4 there is also a zero in $z_i = -1$

$\square$

5) Discrete Fourier Transform (DFT)

The DTFT is a good analytical method to study filters. However computers struggle to directly compute it. In this chapter we will see how the DFT can be calculated faster and still be useful as a sampled version of the DTFT

The Discrete Fourier Transform (DFT) is the Fourier transform relative to **finite length or periodic discrete-time signals**. Periodic and finite time are equivalent since both cases are fully specified by a finite number of samples $n = 0, \dots, N-1$.

> **Definition 12** (Discrete Fourier Transform)**.**
>
> The DFT is given by:
>
> $$X(kF) = \sum_{n=n_0}^{n_0+N-1} Tx[nT]e^{-j2\pi kFnT} \to \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$$
>
> and the Inverse DFT is given by:
>
> $$x[nT] = \sum_{k=k_0}^{k_0+N-1} FX(kF)e^{+j2\pi kFnT} \to \sum_{k=0}^{N-1} X(k)e^{+j\frac{2\pi}{N}kn}$$
>
> The two summations are periodic with period NT and NF respectively. Moreover $TF = 1/N$. We will usually assume $F = 1$ for the DFT and $T = 1$ for the IDFT so that the constant $2\pi kFnT$ becomes $\frac{2\pi}{N}kn$.

The DFT is periodic in $k = \{0, \dots, N-1\}$, that is, it has period N. We well see that the length N is related on how dense the samples are.

Properties of the DFT

Let $x[n], y[n]$ have as DFT $X(z), Y(z)$ and $\alpha, \beta \in \mathbb{C}$ and $x[n]$ with length N

Linearity

$$w[n] = \alpha x[n] + \beta y[n] \overset{\text{DFT}}{\longrightarrow} W(k) = \alpha X(k) + \beta Y(k)$$

Periodicity

$$X(k) = X(k+N)$$

Moreover the IDFT returns a periodic version of $x[n]$ which has the same property

$$x[n] = x[n+N]$$

In the case of zero padding where the DFT has length we have

$$x[n] = x[n+M]$$

Time Shift

$$w[n] = x[n-n_0] \overset{\text{DFT}}{\longrightarrow} W(k) = e^{-j\dfrac{2\pi}{N}kn_0}X(k)$$

Conjugate Symmetry

$$X(k) = X^*(-k)$$

And by the periodicity

$$X(k) = X^*(-k) = X^*(N-k)$$

In the case of zero padding where the DFT has length $M > N$ we have

$$X(k) = X^*(-k) = X^*(M-k)$$

Circular Convolution

$$W(k) = X(k)Y(k) \overset{\text{IDFT}}{\longrightarrow} w[n] = (x * y)[n] = \sum_{l=0}^{N-1} x[l]y[n-l] = \sum_{l=0}^{N-1} x[n-l]y[l]$$

The last, most important property is the **Relation between DFT, DTFT and z-Transform**, which will be explained more in depth in the next chapter. Basically it states that:
The DFT corresponds to a sampled version of the z-Transform on the unit circle. It follows that the DFT is the sampled version of the DTFT of the aperiodic signal at the frequencies $\theta_k = \frac{2\pi}{N}k$.

5.2) Relationship between the DFT and DTFT

The DFT corresponds to a sampled version of the z-Transform on the unit circle. It follows that the DFT is the sampled version of the DTFT of the aperiodic signal at the frequencies $\theta_k = \frac{2\pi}{N}k$.

- If $N = M$ we have the same number of samples of the support of the signal
- If $N > M$ we have a denser set of samples to better approximate the DTFT. This is called **zero padding** since we add some zeroes to the right hand side of the signal.
- If $N < M$ then there is temporal aliasing since not all the samples are included

---

Here is a quick proof:
Consider an aperiodic signal with finite support $\{0, \dots, M-1\}$. It has the following z-Transform

$$x[n] = 0 \; \forall n \notin \{0, .., M-1\} \rightarrow X(z) = \sum_{n=0}^{M-1} x[n]z^{-n}$$

Now let's make this signal applicable to the DFT by making it periodic with a period N of at least M samples: $N \geq M$. That is: $\tilde{x}[n] = rep_N x[n] = \sum_{l=-\infty}^{\infty} x[n-lN]$
By directly applying the DFT we get:

$$\tilde{X}(k) = \sum_{n=0}^{N-1} \tilde{x}[n]e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} x[n]z^{-n} = X(z)|_{z=z_k}$$

By setting $z = z_k = e^{j\frac{2\pi}{N}k}$ and by noticing that in the sum $x[n] = \tilde{x}[n]$ it is clear that the DFT of the periodic signal is the sampling of N roots (N equally spaced points) of the z-Transform of the aperiodic signal on the unit circle. Since on the unit circle the z-Transform equals the DTFT (if it converges), then it is clear how the DFT is a sampled version of the DTFT.

□

A quick explanation of zero padding:
By zero padding we implicitly add zeroes to the signal where we take the DFT.
Let $x[n]$ be non zero in the interval $\{0, \dots, L-1\}$ (length L), the padded signal will be $x_p[n] = \begin{cases} x[n] & 0 \leq n \leq L-1 \\ 0 & L \leq n \leq N-1 \end{cases}$

The zero padded DFT is:

$$X_N[k] = \sum_{n=0}^{N-1} x_p[n]\, e^{-j\frac{2\pi k}{N}n} = \sum_{n=0}^{L-1} x[n]\, e^{-j\frac{2\pi k}{N}n} = X\!\left(e^{j\frac{2\pi}{N}k}\right)$$

where we can change $x_p[n]$ with $x[n]$ by removing the terms that don't contribute to the sum.

Here we see that the DFT is calculated using the non padded signal, thus still from 0 to L-1, but with denser samples on the z-transform. The samples are always taken in the interval $[0, 2\pi)$ with steps of $\Delta\omega = 2\pi/N$.

Since k is mapped to the frequencies using $\theta = \frac{2\pi}{N}k$ changing N also changes the map and thus even though in the summation the term $\frac{2\pi}{N}$ is different also the frequency related to it is different and the DFT is still correct, only sampled at different points. Here is an example:
A signal of length L and DFT of length $N = L$ will have at frequency π, that is $k = L/2$. Here the complex exponential will have $\frac{2\pi}{N}k = \pi$. With DFT of length $N' = 2L$ frequency π will be at $k' = L$ and thus $\frac{2\pi}{N'}k' = \pi$ still.

5.3) Fast Fourier Transform (FFT)

The FFT refers to the class of algorithms used to efficiently compute the DFT. This algorithm allows to compute the DFT in $O(N\log_2 N)$ instead of $O(N^2)$. This chapter will be a quick rundown of the algorithm.

For simplicity consider $N = 2^v,\ v \in \mathbb{N}$ to easier see the **divide and conquer** approach.

Let's start with some notation:
We denote $W_N = e^{-j\frac{2\pi}{N}}$ a particular root of the unit circle. Here the DFT and IDFT become:

$$X(k) = \sum_{n=0}^{N-1} x[n]{W_N}^{kn}$$
$$x[n] = \sum_{k=0}^{N-1} X(k){W_N}^{-kn}$$

The direct computation of ONE coefficient requires:

- N complex multiplications
- N-1 Complex additions
- N+1 memory access operations to read $x[0], x[N-1]$ and to write $X(k)$
- N memory access to read the N values of the coefficients
  This is $O(N)$ and therefore to compute all N values we have $T_D(N) = O(N^2)$.

By dividing the sequence in two parts, the total time complexity becomes $T_F = 2T_D(N/2) + R(N)$. By iterating until we have just two samples we reach the total time of $T_F(N) = O(N\log_2 N)$

Why?
The decimation in time divides first from $x_0, \dots, x_{N-1}$ to e sequence of even indexes and one of odd ones. Now the DFT can be written as:

$$\begin{aligned}
X(k) &= \sum_{r=0}^{\frac{N}{2}-1} x[2r]{W_N}^{2rk} + \sum_{r=0}^{\frac{N}{2}-1} x[2r+1]{W_N}^{(2r+1)k} \\
&= \sum_{r=0}^{\frac{N}{2}-1} x[2r]{W_{\frac{N}{2}}}^{2rk} + {W_N}^{k} \sum_{r=0}^{\frac{N}{2}-1} x[2r+1]{W_{\frac{N}{2}}}^{(2r)k} \\
&= G(k) + {W_N}^k H(k)
\end{aligned}$$

Where we used ${W_N}^2 = e^{-j2\frac{2\pi}{N}} = W_{N/2}$

Here G and H are two DFT on N/2 points and therefore have period N/2.
The recombination cost is proportional to N and therefore we now have $2(N/2)^2 + N < N^2$ already a progress!

By exploiting the periodicity we have that $X(k = \{0, .., \frac{N}{2} - 1\}) = G(k) + W_N^k H(k)$ and the other half points have $X(\frac{N}{2} + k = \{0, \dots, \frac{N}{2} - 1\}) = G(k) - W_N^k H(k)$. This was done by noticing that

${W_N}^{\frac{N}{2}+k} = {W_N}^{\frac{N}{2}}{W_N}^{k} = -W_N^k$. And thus we need N/2 multiplications and N sums.

Remark.

In this chapter we discussed the DFT of a signal. Here we have that $N \geq D_X$ in order to avoid time aliasing.

In the next chapters we will see the DFT of a system (convolution). Here the length of the convolution will be $D_y = L + M - 1$ and therefore $N \geq D_y = L + M - 1$

5.4) DFT And Convolutions

One property of the DFT is the **circular convolution**

Theorem 13 (Circular Convolution)**.**

Let $x[n]$ and $h[n]$ have the following DFT $X(k)$ and $H(k)$, then the IDFT of $Y(k) = X(k)H(k)$ is:

$$y[n] = (x * h)[n] = \sum_{l=0}^{N-1} x[l]h[n-l]$$

Which differs from the IDTFT of the convolution since in that case we would have the classic linear convolution

$$y[n] = (x * h)[n] = \sum_{l=-\infty}^{\infty} x[l]h[n-l]$$

To do the frequency analysis using the DFT we need the linear convolution. Let's see under what circumstances the two are equal.

$D_x \approx M$	$D_x >> M$
FFT approach when $M \geq 2^4$	FFT approach when $D_x < 2^{M/3} \leftrightarrow M > 3\log_2 D_x$

Here is a quick proof:
Consider the periodic signals and their DFT

$$\tilde{x}[n] = rep_N x[n] \to \tilde{X}(k)$$
$$\tilde{h}[n] = rep_N h[n] \to \tilde{H}(k)$$

The product of the DFT is the *circular* convolution between the two functions.
Let $x[n]$ have finite length D_X and $h[n]$ the impulse response of a FIR of length M, then also the *linear* convolution $y[n] = (x * h)[n]$ has limited length $D_y = D_X + M - 1$. As already seen here.

If $N \geq D_y$ there is no time aliasing as seen here. And the two convolutions are equal for $0 \leq n \leq N-1$. Note that for efficiency it is useful to choose N a power of two: $N = \min\{2^v : 2^v \geq D_y\}$.

- In time we need to compute M multiplications and M-1 additions, that is $T_D = K_D M$

- In frequency we need two FFT, one IFFT and N multiplications. The time per sample is:
  $T_F = \frac{1}{N}(3T_{FFT+NK_{mult}} = \frac{1}{N}(3K_F N \log_2 N + NK_{mult} \approx 3K_F \log_2 N = O(\log_2 N)$

The ratio is:

$$\frac{T_F}{T_D} \approx \frac{3K_F \log_2 N}{K_D M}$$

This favors the FFT approach when the ratio is smaller than one.

Two cases occur: by supposing $K_F = K_D$

1. $D_X \approx M$
   In this case we have $N = D_x + M - 1 \approx 2M$:
   $$\frac{T_F}{T_D} \approx \frac{3\log_2 2M}{M} = 3\frac{1+\log_2 M}{M}$$
   Which is more efficient for the FFT approach when $M \geq 2^4$
2. $D_x >> M$
   In this case $N \approx D_x$:
   $$\frac{T_F}{T_D} \approx \frac{3\log_2 D_x}{M}$$
   Which is more efficient for the FFT approach when $D_x < 2^{M/3} \to M > 3\log_2 D_x$

$\square$

5.4.1) Block Convolution

To filter a signal of arbitrary length with a FIR of length M we can compute $y[n]$ by dividing the signal into blocks and convoluting them with $h[n]$ and then appropriately combining them.
These following methods have same computational complexity:

Overlap-Add Method

Here we split the signal into non overlapping blocks of size $L > M$. The blocks are given by

$$x_k[n] = \begin{cases} x[n] & kL \leq n \leq (k+1)L - 1 \\ 0 & \text{elsewhere} \end{cases} = x[n]w[n-kL]$$

Where the window is defined as $w[n] = u[n] - u[n-L]$ that is $w[n-kL] = u[n-kL] - u[n-k]$.

Assuming that $x[n]$ is causal we have

$$y[n] = \left(\sum_{k=0}^{\infty} x_k[n]\right) * h[n] = \sum_{k=0}^{\infty} y_k[n]$$

Where each component of y_k has length $L + M - 1$ and support $\{kL, \dots, (k+1)L + M - 2\}$
Between y_k and y_{k+1} there is an overlap of M-1 samples since

$$\begin{matrix} y_k = \{kL, .., (k+1)L + M - 2\} \\ y_{k+1} = \{(k+1)L, \dots, (k+2)L + M - 2)\} \end{matrix} \xrightarrow{\cap} \{(k+1)L, .., (k+1)L + M - 2\}$$

The overlap has length $M - 2$ and thus $M - 1$ samples that overlap in the support, therefore we will have more than one contribution:

$$y[n] = \sum_{k=0}^{+\infty} y_k[n] = \begin{cases} y_{k-1}[n] + y_k[n] & kL \leq n \leq kL + M - 2 \\ y_k[n] & kL + M - 2 < n < (k+1)L \\ y_k[n] + y_{k+1}[n] & (k+1)L \leq n < (k+1)L + M - 2 \end{cases}$$

Formula

We can compute this with a DFT of length $N \geq L + M - 1$ and it was found that we need at least $N = 10^3$

Overlap Save Method

Here the windows are overlapped by $M-1$ samples. Then, only the portion of the values of the partial convolutions $y_k[n]$ that are coincident with the convolution $y[n]$ are considered to provide the final result.

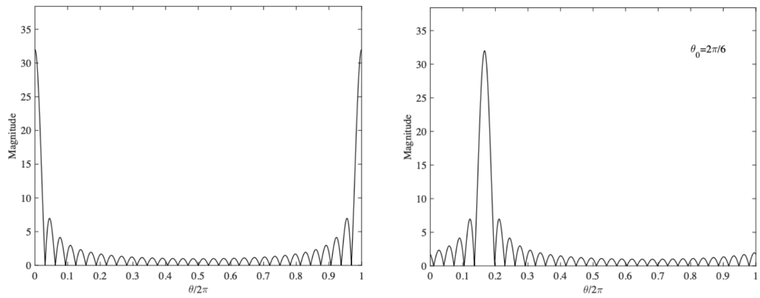
5.5) Windowing

The frequency analysis of a windowed segment of the signal differs from the non windowed case.

First of all recall that in windowing we have access only to a windowed segment of the signal, therefore we must consider the transform of that window.
Let $y[n] = x[n]w_r[n]$ with $x[n] = Ae^{j\theta_0 n}$ we obtain:

$$Y(e^{j\theta}) = AW_R(e^{j(\theta-\theta_0)})$$
$$\tilde{Y}(f) = AW_R(f-f_0)$$

By recalling that $w_r[n]$ is a periodic pulse we know that $W_R(f)$ is a sinc function that is now centered in f_0 (or θ_0). Therefore the main lobe will be proportional to $|A|$ and centered at the frequency of the signal.

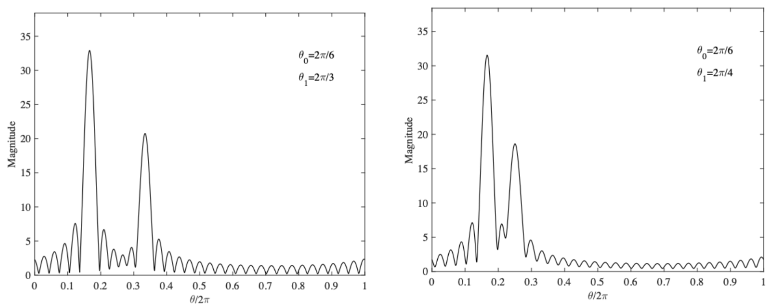


Magnitude of $W_R(e^{j\theta})$ and of $Y(e^{j\theta}) = AW_R(e^{j(\theta-\theta_0)})$ with $N=32, A=1, \theta_0=\frac{2\pi}{6}$

Example

If there are more complex exponentials present in the input we will end up with 2 peaks due to superimposition. We call **leakage** the phenomenon that changes the peak of one exponential due to the contribution of the secondary peaks of the other exponentials.

$$x[n] = A_0 e^{j\theta_0 n} + A_1 e^{j\theta_1 n} \xrightarrow{\text{DTFT}} Y(e^{j\theta}) = X(e^{j\theta}) * W_R(e^{j\theta}) = A_0 W_R(e^{j(\theta-\theta_0)}) + A_1 W_R(e^{j(\theta-\theta_1)})$$



Example of Leakage

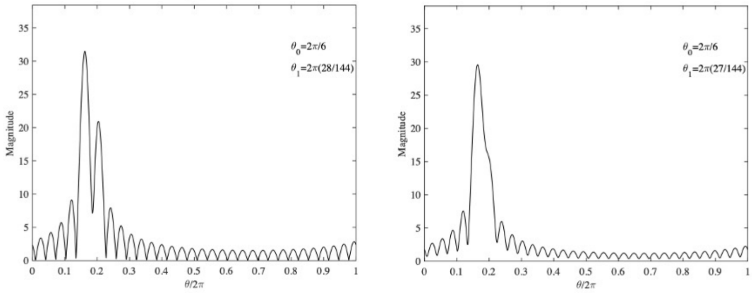
> **Remark 14.**
>
> If the exponentials are close in frequency they might be undistinguishable in the windowed DTFT

To have a better resolution recall this

> **Theorem 15.**

The number of side lobes visible in a DTFT plot increases ***linearly*** *with* L

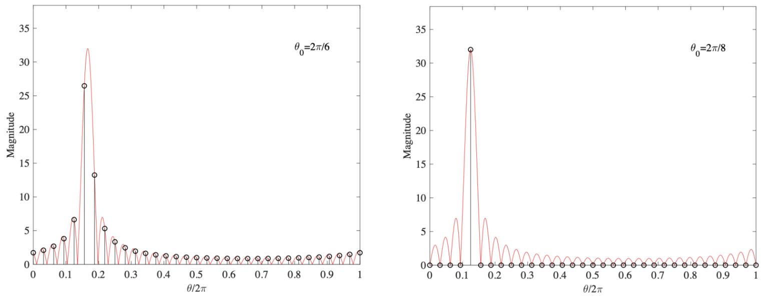
Therefore the more frequency there are it is better to use higher L values for the window to avoid leaking as seen in the following graphs.



Magnitude of $Y(e^{j\theta}) = AW_R(e^{j(\theta-\theta_0)})$ *with* $A_0 = 1, A_1 = 0.6$

Effect of Higher N on Leaking

However computers can calculate the DFT and therefore the samples $\theta_k = \frac{2\pi}{N}k$ must contain the frequencies of the main lobes, otherwise the DFT plot will lead to inaccurate results.



Effect of sampling the DTFT when computing the DFT.
Plot of $Y(e^{j\theta}) = AW_R(e^{j(\theta-\theta_0)})$ with $N = 32$ and different values of θ_0.

Example

To avoid resolution errors it is possible to use zero padding and with a higher N for the DFT we end up with denser samples.

Remark.

By increasing the window size we end up with more and better separated lobes. We end up with

- Main lobe of width $4\pi/L$
- Side lobes spacing of $2\pi/L$

The size of the DFT indicates how many samples are used for the DFT calculation. If $N > L$ we have zero padding.

Remark (Fourier Analysis of Sinusoidal Signal)**.**

A sinusoidal signal can be expressed as a sum of complex exponentials, but we actually only need to estimate one of the two since the second one is obtained by symmetry.

5.6) Spectrogram

What happens if we have an indefinitely long non-periodic signal?

- The signal has the same global spectral content, the DTFT exists but the DFT computation might take a long time
- The signal is a concatenation of smaller signals with different local spectral content, this is a collection of many short length DFTs

We will develop the **spectrogram:** a tool to see the spectral content change over time

To see the temporal frequency changes in the long signal we introduce the **Short Time Discrete Fourier Transform (STDFT)** and is defined as:

Definition 16 (Short Time Discrete Fourier Transform)**.**

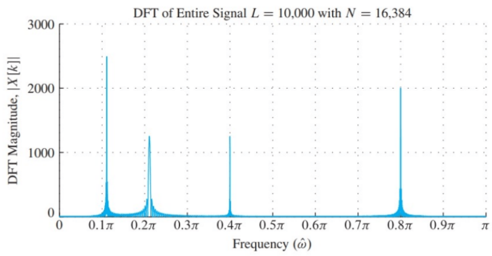
$$X[k, n_s] = \sum_{m=0}^{L-1} w[m]x[n_S + m]e^{-j(\frac{2\pi k}{N})m}$$

This transform involves 2 steps:

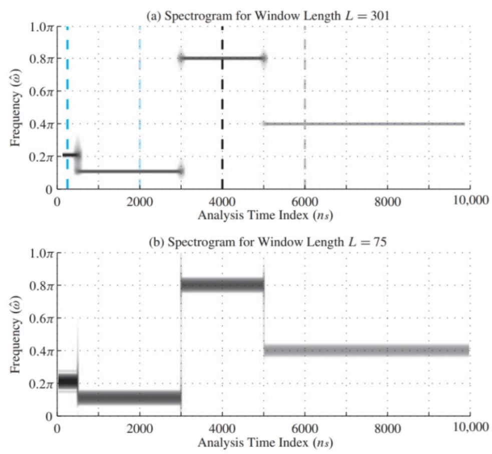
- **windowing:** $w[n]$ is a windowing sequence that is nonzero only in $\{0, \ldots, L-1\}$ with $L << D_x$. The window selects a finite segment of the signal
- **short length DFT:** extracts the spectrum (DFT) of the short interval

Usually the time n_s is advanced in steps of $1 \leq R < L$ samples.

Take for example the following DFT:



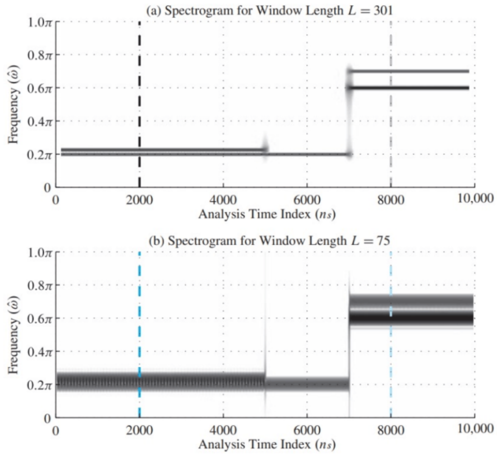
DFT



Spectrogram Representations With Different Windows

Higher values of L will cause more aliasing in the temporal changes location

Not all signals have only one frequency at any point in time. In that case resolution is also important, since close frequencies can mix into one another.



Resolution Difference

Here we can see how a bigger window gives us a better frequency resolution. Near frequencies might overlap as seen already with the DFT windowing. In a Hann window we can find $\Delta\hat{\omega} \approx C/L$ with $4\pi < C < 8\pi$. If two frequencies differ by less than $\Delta\hat{\omega}$ then their STDFT peaks will blend together

Finally we can recap the window size as:

Window Size	**Time Resolution**	**Frequency Resolution**
Big	Worse	Better
Small	Better	Worse

Remark.

A denser DFT does not mean better frequency resolution, this is given by L since the width of the main lobes is of $4\pi/L$

6) Design of (Low-Pass, Type 1) Linear Phase Filter

Physically realizable filters must be causal and finite, however ideal filters cannot satisfy these requirements.

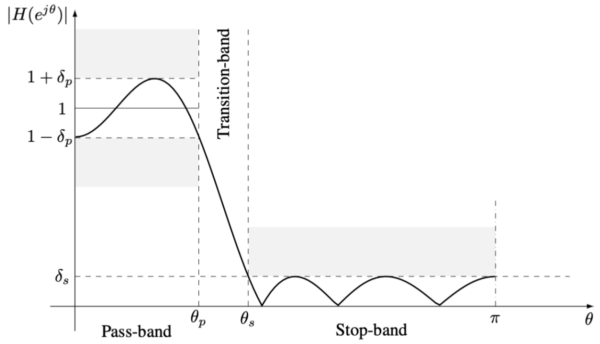
An ideal low pass filter will have frequency response and IDTFT of:

$$\mathcal{H}e^{j\omega} = \text{rect}\left(\frac{\omega}{2\omega_c}\right) \overset{\text{IDTFT}}{\rightarrow} h[n] = \begin{cases} \text{sinc}\left(\frac{\omega_c}{\pi}n\right) & n \neq 0 \\ \frac{\omega_c}{n} & n = 0 \end{cases}$$

here it is clear that $h[n]$ is both non causal and infinite.

A real low pass filter is distinguished by three zones: pass, transition and stop band
And there are 4 main design parameters:

- δ_p: pass-band error
- δ_s: stop-band error
- θ_p: pass-band frequency limit
- θ_s: stop-band frequency limit



Example

Usually these values are given in dB: Pass-Band Ripple (PBR) and Stop-Band Attenuation (A):

$$PBR_{dB} = 20\log_{10}\frac{\max\limits_{\theta\in[0,\theta_p]}|H(\theta)|}{\min\limits_{\theta\in[0,\theta_p]}|H(\theta|} = 20\log_{10}\frac{1+\delta_p}{1-\delta_p}$$

$$A_{dB} = -20\log_{10}\max_{\theta\in[\theta_s,\pi]}|H(\theta)| = -20\log_{10}\delta_s$$

Recall that a linear FIR filter has symmetric or antisymmetric impulse response and that it's frequency response can be written as $H(\theta) = e^{j\beta}e^{-j\frac{N}{2}\theta}Q(\theta)P(\theta)$. Where in particular the type 1 linear phase FIR filter has $\beta = 0,\ Q(\theta) = 1,\ N$ even , $r = \frac{N}{2}$ $p_0 = h[\frac{N}{2}]$ $p_n = 2h[\frac{N}{2} - n]$ $n \in \{1, .., \frac{N}{2}\}$. And thus:

$$\underset{\text{Type 1 FIR}}{H(\theta)} = e^{-j\frac{N}{2}\theta}\sum_{n=0}^{\frac{N}{2}} 2h[\frac{N}{2} - n]\cos(n\theta)$$

While our target function is

$$\underset{\text{target}}{H(\theta)} = e^{-j\frac{N}{2}\theta}D(\theta)$$

With $D(\theta)$ a non-negative, real periodic function with period 2π that represents the desired behaviour of our target. Moreover since it is periodic it allows Fourier series expansion and coefficients.

We will study 3 main ways to design a filter.

6.1) Windowing Technique

The windowing technique consists in using an ideal impulse response $h_d[n]$ for the filter and multiplying it with a window of length L in order to make it realizable:

$$h[n] = \overline{h}[n - \frac{N}{2}] \qquad \overline{h}[n] = h_d[n]w_R[n]$$

With a rectangular window, as L increases we obtain a better approximation at the constant points of the filter but more oscillations at the points of discontinuity. These errors are around a 9% $\implies \delta_p = \delta_s = 0.09$ and cannot be reduced. Moreover, as explained below the main lobe will dictate the width of the transition band of $\Delta\theta_{p/s} = \frac{2\pi}{L}$

Pros	Cons
very simple and intuitive	not optimal results
maintains zero coefficients of ideal impulse response, useful for interpolation (Nyquist filters)	Same band error

Moreover different types of windows allow for a better approximation. All these windows will have these characteristics:

- Width of transition band proportional of reciprocal of L: $\frac{\theta_s - \theta_p}{2\pi} \approx \frac{\alpha}{L}$
- The errors are the same: $\delta_s \approx \delta_p$
- Both previous values are independent of the cutoff frequency $\theta_0 = \frac{\theta_s + \theta_p}{2}$

Window type	Main lobe width	Maximum side lobe amplitude	Transition bandwidth	Minimum stop-band attenuation
Rectangular	$2\frac{2\pi}{L}$	-13.3 dB	$0.9\frac{2\pi}{L}$	21 dB
Triangular	$4\frac{2\pi}{L}$	-26.5 dB	$3\frac{2\pi}{L}$	26 dB
Hanning	$4\frac{2\pi}{L}$	-31.5 dB	$3.1\frac{2\pi}{L}$	44 dB
Hamming	$4\frac{2\pi}{L}$	-42.7 dB	$3.3\frac{2\pi}{L}$	54 dB
Blackman	$6\frac{2\pi}{L}$	-57 dB	$5.5\frac{2\pi}{L}$	74 dB

Table

- Rectangular window:

$$w_R[n] = \begin{cases} 1 & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- Triangular/Bartlett window: This is the convolution of two rects, less error from side lobes, but more width

$$w[n] = \begin{cases} 1 - \frac{|n|}{N} & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- Hanning window: raised cosine with 100% roll-off

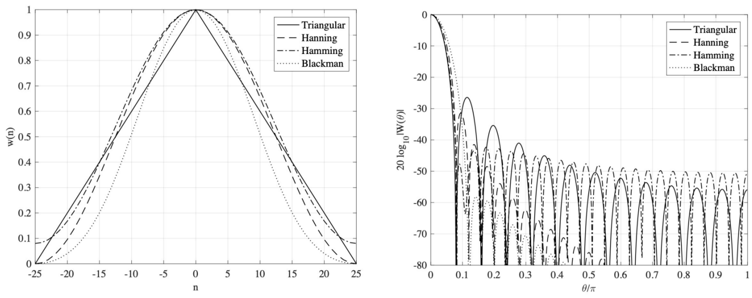
$$w[n] = \begin{cases} \frac{1}{2} + \frac{1}{2}\cos\left(\frac{\pi n}{\frac{N}{2}}\right) & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- Hamming window: modified raised cosine with 100% roll-off (most used)

$$w[n] = \begin{cases} 0.54 + 0.46\cos\left(\frac{\pi n}{\frac{N}{2}}\right) & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

- Blackman window: modified raised cosine with additional correction term

$$w[n] = \begin{cases} 0.42 + 0.5\cos\left(\frac{\pi n}{\frac{N}{2}}\right) + 0.08\cos\left(\frac{2\pi n}{\frac{N}{2}}\right) & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$



Comparison

The last window is the Kaiser window (also most used) which is a bit more complex

$$w[n] = \begin{cases} \frac{I_0\left(\beta\sqrt{1-(\frac{2n}{N})^2}\right)}{I_0(\beta)} & |n| \leq \frac{N}{2} \\ 0 & \text{elsewhere} \end{cases}$$

Where $I_0(x)$ is the **Bessel function** of the first kind of order 0

$$I_o(x) = 1 + \sum_{l=1}^{\infty} \frac{\frac{x}{2}^l}{l!}$$

The parameter β is used to control the attenuation of the stop band at the expense of the transition band:

$$\beta = \begin{cases} 0.1102(A_{dB} - 8.7) & A_{dB} < 50dB \\ 0.5842(A_{dB} - 21)^{0.4} + 0.07886 & 21dB \leq A_{dB} \leq 50dB \\ 0 & A_{dB} \leq 21 \end{cases}$$

Once β is found we can find N via the empirical formula

$$N \approx \frac{A_{dB} - 8}{14.36\frac{\theta_s - \theta_p}{2\pi}}$$

Here is a quick rundown:

Call $\overline{h}[n] = h[\frac{N}{2} + n]$ then we have

$$\overline{H}(\theta) = \sum_{n=-r}^{r} \overline{h}[n]e^{-jn\theta} = \sum_{n=0}^{r} p_n \cos(n\theta)$$

with $r = -\frac{N}{2},\ p_0 = \overline{h}[n],\ p_n = 2\overline{h}[n]$

Here it is clear that $\overline{h}[n] \approx h_d[n]$ in order to have a low pass filter.

By using the windowing technique we define a rectangular window of length $L = 2r + 1 = N + 1$ an set

$$\overline{h}[n] = h_d[n]w_R[n]$$

so to obtain a non causal response but we just delay it by $r = N/2$ samples so to make it causal.

$$h[n] = \overline{h}[n - \frac{N}{2}] \rightarrow H(z) = z^{-\frac{N}{2}}\overline{H}(z) \rightarrow H(\theta) = e^{-j\frac{N}{2}\theta}\overline{H}(\theta)$$

Since $\overline{h}[n]$ is even symmetric we satisfy $h[n] = h[N - n]$

□

Example.

Suppose to want to design a low pass filter that approximates $D(\theta) = \begin{cases} 1 & -\theta_0 \leq \theta \leq \theta_0 \\ 0 & \text{elsewhere} \end{cases}$

The Fourier coefficients become

$$h_d[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} D(\theta)e^{jn\theta}d\theta = \frac{1}{2\pi}\int_{-\theta_0}^{\theta_0} e^{jn\theta} = \frac{\sin(n\theta)}{n\pi} = \frac{\theta_0}{\pi}\text{sinc}(n\frac{\theta_0}{\pi})$$

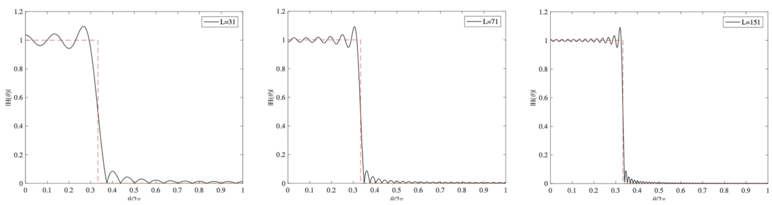
We have a maximum that is less than 1 in $\overline{h}[n] = h[\frac{N}{2}] = h_d[0] = \frac{\theta_0}{\pi}$. *By the windowing technique gives the following result:*

$$h[n] = h_d[n - \frac{N}{2}]w_r[n - \frac{N}{2}] \rightarrow H(\theta) = D(\theta) * W_R(\theta)$$

What happens is that $W_R(\theta)$ *has a max at the origin with height L and width* $4\pi/$ *and zero at* $\theta_k = \frac{2\pi}{L}k$. *Which is a complex calculation since it is a convolution between a rect and a DFT transform of a rect.*

However by using matlab we get the following results:

- *If L is sufficiently large the constant regions of* $D(\theta)$ *are correctly approximated*
- *The* ***Gibbs phenomenon*** *creates oscillating behaviour in points of discontinuity*



Example

6.2) Frequency Sampling Method

In this case we sample the desired frequency response $H_d(\theta)$ at points $\theta_k = \frac{2\pi}{L}k$. From here we obtain the DFT $H(k)$ and by computing the IDFT we obtain the physically realizable $h[n]$. This result is not optimal as it **guarantees correct response only at frequencies** θ_k

We notice that the obtained response is given with the following relation with the ideal response:

$$h[n] = \text{rep}_L h_d[n]$$

with $L = N + 1$

Example.

Suppose we have $H_d(\theta) = e^{-j\frac{N}{2}\theta}D(\theta)$ *at cutoff frequency* $\theta_0 = \frac{\pi}{2}$, *so that* $D(\theta) = \begin{cases} 1 & -\theta_0 \leq \theta \leq \theta_0 \\ 0 & \text{elsewhere} \end{cases}$

We compute the DFT at the $L = N + 1$ *samples* $\theta_k = \frac{2\pi}{N+1}k$ *so that* $H(k) = H_d(\theta_k)$.

6.3) MinMax Method

This method provides the optimal results as **it minimizes the maximum of the approx error** of the response in the pass band and stop band.

Let $D(\theta)$ be the desired frequency response magnitude and $W(\theta)$ a positive weighing function. Both these are continuous on a compact subset $\mathcal{J}$ that is closed an limited of the interval $[0, \pi]$. We want to find $\overline{H}(\theta)$ that minimizes the following:

$$E(\theta) = W(\theta)(D(\theta) - \overline{H}(\theta))$$

that is

$$\min_h ||E(\theta)||_\infty = \min_h \max_{\theta \in \mathcal{J}} |W(\theta)(D(\theta) - \overline{H}(\theta))|$$

Remark.

For a low pass filter with pass band in $[0, \theta_p]$ and stop band in $[\theta_s, \pi]$ we define

$$\mathcal{J} = [0, \theta_p] \cup [\theta_s, \pi]$$

An alternate formulation is done by considering $\overline{H}$ as a linear combination of $r+1$ sinusoids

$$\overline{H}(\theta) = \sum_{n=0}^{r} p_n \cos(n\theta)$$

the the necessary and sufficient condition for the minmax problem is that $E(\theta)$ presents at least $r+2$ **alternations**, that are points where

$$\theta_i \in \mathcal{J} : E(\theta_i) = -E(\theta_i)$$

Alternation points are boundary points of the compact subset $\mathcal{J}$ or points where the derivative of $E(\theta)$ is zero, i.e., points of local maxima or minima of $E(\theta)$.

An algorithm for the solution of this problem was found by Remez. If the alternation points are known, then the following $r+2$ equations hold

$$W(\theta_i)\left(D(\theta_i) - \sum_{0}^{r} p_n \cos(n\theta_i)\right) = (-1)^i \delta \quad i = \{0, .., r+1\}$$

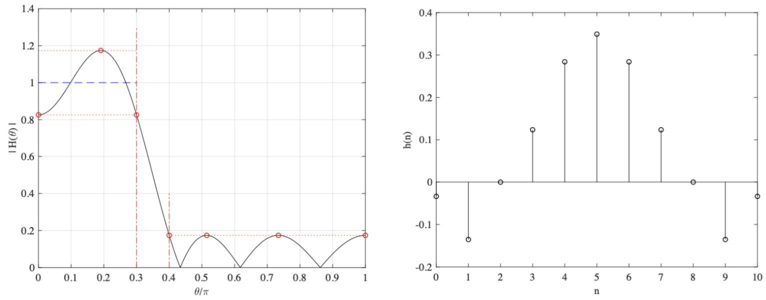
This is a system of $r+2$ equations and unknowns p_n and another unknown δ where $|\delta| = ||E(\theta)||_\infty$
The algorithm first guesses the alternation points and at each iteration it updates the alternation points until the optimal solution is found.

Usually we use a constant weighing function $W(\theta)$. If the errors of the two band are different, then we can define $W(\theta)$ as two different functions.
Let $\delta_p = A\delta_s$, then $w_p = \frac{\delta_s}{\delta_p} w_s = w_s/A$

$$W(\theta) = \begin{cases} w_p = \frac{1}{A} & \theta \in [0, \theta_p] \\ w_s = 1 & \theta \in [\theta_s, \pi] \end{cases}$$

Example.

Here we have $\mathcal{J} = [0, 0.3\pi] \cap [0.4\pi, \pi]$. *With a Type 1 FIR filter of length* $N = 10$ *we end up with* $r + 1 = \frac{N}{2} + 1 = 6$ *degrees of freedom and therefore* $r + 1 = 7$ *alternation points as seen in the graph.*



Example

6.4) Extension to the Design of a Generic Linear-Phase FIR filter

Recall how any FIR filter has amplitude response

$$H(\theta) = Q(\theta)P(\theta)$$

Then we apply the alternation theorem

$$\begin{aligned} E(\theta) &= W(D - \overline{H}) = W(\theta)(D(\theta) - Q(\theta)P(\theta)) \\ &= W(\theta)Q(\theta)\left(\frac{D(\theta)}{Q(\theta)} - P(\theta)\right) \\ &= \hat{W}(\theta)(\hat{D}(\theta) - \overline{H}(\theta)) \end{aligned}$$

So the theorem holds with a normalized weighing function and amplitude function

In any case the **filter order** is approximately:

$$N \approx \frac{-10\log_{10}(\delta_p \delta_s) - 13}{14.6\frac{\theta_s - \theta_p}{2\pi}}$$

Remark.

The filter order N is inversely proportional to the transition bandwidth

The order depends on the product $\delta_s \delta_p$ and thus same orders will go with different pairs that allow same product

Band pass/stop filters have two transition bands. Then after computing N_1, N_2 for the two band we have that $\max N_1, N_2 \leq N \leq N_1 + N_2$

7) Multirate Signal Processing

Multirate systems deal with signals that have different sampling frequencies. They are usually used to convert from one sampling frequency to another. A working assumption in this case is that the sampled signal are sampled under Nyquist conditions.

- **Interpolation:** $F_S' = LF_S, L \in \mathbb{N}^+$
- **Decimation:** $F_S' = \frac{1}{M}F_S, M \in \mathbb{N}^+$
- **Generic:** $F_S' = \frac{L}{M}F_S$

7.1) Interpolation

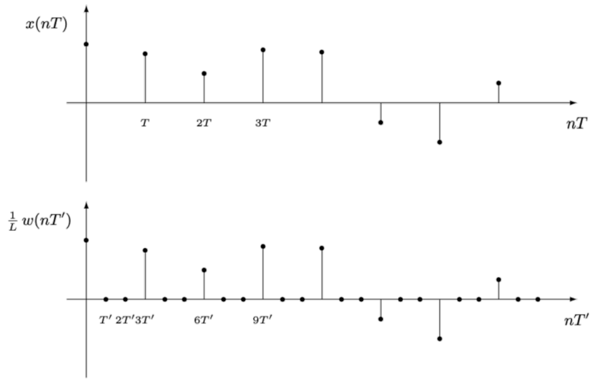
The idea behind interpolation is to add $L-1$ new samples between two adjacent samples. We define the new frequency and period as:

$$F_S' = LF_S \qquad T' = \frac{1}{L}T \qquad L \in \mathbb{N}^+$$

The most basic implementation is to set these samples to 0, that is:

$$w[kT'] = \begin{cases} Lx[nT] & kT' = nT \leftrightarrow k = nL \\ 0 & kT' \neq nT \end{cases}$$

This is LTI. Notice that the filter preserves the information of the original signal. The L factor scales the original samples but this preserves both the average energy and allows for a correct DFT since the zero samples added will cause a scaling error of 1/L.



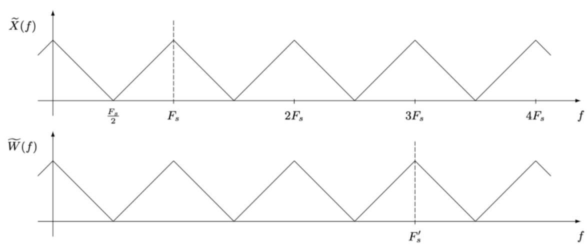
Example

Then, after interpolating, we will add a **LPF with cutoff frequency $\mathbf{F_s/2}$**

In fact, let's compute the DFT:

$$\tilde{W}(f) = T' \sum_{k=-\infty}^{\infty} w[kT']e^{-j2\pi fkT'} = T' \sum_{n=-\infty}^{\infty} Lx[nT]e^{-j2\pi fnLT'} \overset{T'=LT}{=} T \sum_{k=-\infty}^{\infty} x[nT]e^{-j2\pi nfT} = \tilde{X}(f)$$

The the second summation was obtained by recalling that $w[kT'] = 0$ unless $k = nL$. If we only take these terms we can reach the form of the second summation. Finally we recognize that these are equivalent but with different periods.



Example with L=3

Let's Compute the z-Transform:

$$W(z) = T' \sum_{k=-\infty}^{\infty} w[kT']z^{-k} = T' \sum_{n=-\infty}^{\infty} Lx[nT]z^{-nL} \overset{T'=LT}{=} T \sum_{n=-\infty}^{\infty} x[nT](z^L)^{-n} = X(z^L)$$

Here the steps are roughly the same but yield a result that replicates the Z transform L times in frequency

Therefore the original DFT has frequencies in

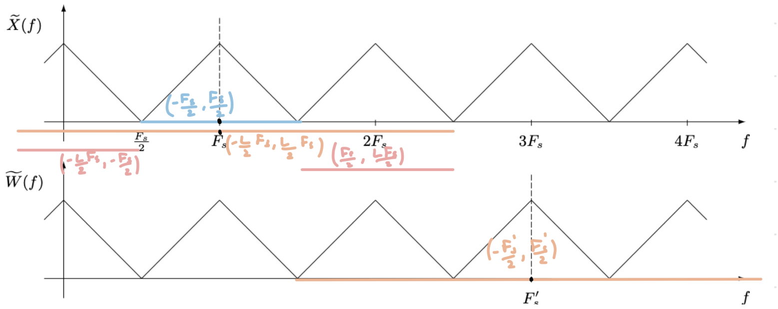
$$\left(-\frac{1}{2}F_S, \frac{1}{2}F_S\right)$$

To correctly preserve the information notice that w contains the frequencies of the signal in

$$\left(-\frac{1}{2}F_S', \frac{1}{2}F_S'\right) = \left(-\frac{L}{2}F_S, \frac{L}{2}F_S\right)$$

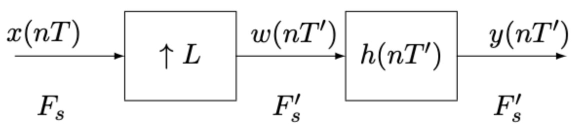
Therefore we need to remove the frequencies in

$$\left(-\frac{1}{2}F_s', -\frac{1}{2}F_S\right) \cap \left(\frac{1}{2}F_s, \frac{1}{2}F_S'\right) = \left(-\frac{L}{2}F_s, -\frac{1}{2}F_S\right) \cap \left(\frac{L}{2}F_s, \frac{1}{2}F_S\right)$$



Colored Version Of Frequencies

This can be achieved via an ideal low pass filter with cutoff frequency at $f_0 = \frac{1}{2}F_S$



Complete System

$$y[nT'] = T \sum_{k=-\infty}^{\infty} x[kT]h[nT' - kT']$$
$$Y(z) = H(z)W(z) = H(z)X(z^L)$$
$$\tilde{Y}(f) = \tilde{H}(f)\tilde{W}(f) = \tilde{H}(f)\tilde{X}(f)$$

$$h[nT'] = F_S \operatorname{sinc}\left(\frac{n}{L}\right)$$

Linear Interpolation

Linear interpolation adds $L-1$ new samples but puts them on a straight line that connects the two adjacent points. It uses the same interpolation system as before but with a different $h[n]$.

Recall that, with same frequency, if we set $x[n] = \delta[n]$ then the system has it's impulse response as output. Recall also that the discrete dirac delta with set period is defined as:

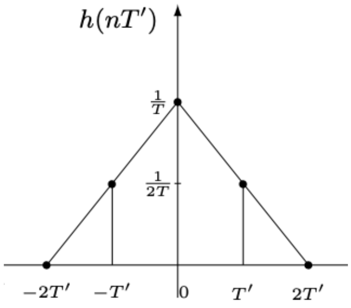
$$\delta[nT] = \begin{cases} \frac{1}{T} & n = 0 \\ 0 & n \neq 0 \end{cases}$$

Let's input it to our system to show that the relationship holds:

$$y[nT'] \overset{x[nT]=\delta[nT]}{=} T \sum_{k=-\infty}^{\infty} \delta[nT]h[nT' - kT'] = T\frac{1}{T}h[nT'] = h[nT']$$

Now it is possible to find the filter with an easy geometric proof since the output of the system $y[nT', x[nT] = \delta[nT]] = h[nT']$ is a triangle with vertices: $(-L, 0),\ (L, 0),\ (0, 1/T)$. And therefore finally

$$h[nT'] = \begin{cases} \frac{1}{T}(1 - \frac{|n|}{L}) & n < L \\ 0 & \text{elsewhere} \end{cases}$$



Example h[nT] with L=2

now let's compute the DFT (case $L = 2$)

$$\begin{aligned} \tilde{H}(f) &= T' \sum_{n=-\infty}^{\infty} h[nT'] e^{-j2\pi f n T'} = \frac{T'}{T} \sum_{n=-1}^{1} (1 - \frac{|n|}{2}) e^{-j2\pi f n T'} \\ &= \frac{1}{2}\left(\frac{1}{2T} e^{j2\pi f T'} + \frac{1}{T} + \frac{1}{2T} e^{-j2\pi f T'}\right) \\ &= \frac{1}{2}(1 + \cos(2\pi f T')) \end{aligned}$$

Exercise.

Now compute the DFT in the generic case:

$$\tilde{H}(f) = T' \sum_{n=-\infty}^{\infty} h[nT'] e^{-j2\pi f n T'} = \frac{T'}{T} \sum_{n=-L+1}^{L-1} (1 - \frac{|n|}{L}) e^{-j2\pi f n T'}$$

Now let $M = L - 1$ and $\alpha = 2\pi f T'$ and recall $T'/T = 1/L$

$$= \frac{1}{L}\left(\sum_{n=-M}^{M} e^{-j\alpha n} - \frac{1}{L} \sum_{n=-M}^{M} |n| e^{-j\alpha n}\right)$$

The first term is the Dirichlet form

$$\begin{aligned} \sum_{n=-M}^{M} e^{-j\alpha n} &= e^0 + \sum_{n=-M}^{-1} e^{-j\alpha n} + \sum_{n=1}^{M} e^{-j\alpha n} = 1 + \sum_{n=1}^{M} e^{j\alpha n} + e^{-j\alpha n} \\ &= 1 + 2\sum_{n=1}^{M} \cos(\alpha n) \end{aligned}$$

And the second term is

$$\begin{aligned} \sum_{n=-M}^{M} |n| e^{-j\alpha n} &= \sum_{n=-M}^{-1} |n| e^{-j\alpha n} + \sum_{n=1}^{M} |n| e^{-j\alpha n} = \sum_{n=1}^{M} n(e^{j\alpha n} + e^{-j\alpha n}) \\ &= 2\sum_{n=1}^{M} n \cos(\alpha n) \end{aligned}$$

Finally we have

$$\tilde{H}(f) = \frac{1}{L}\left(1 + 2(1 - \frac{1}{L})\sum_{n=1}^{L-1}\cos(2\pi f n T')\right)$$

Evaluation of Distortion in Non Ideal Filters

From the results obtained before, an interpolator must have an ideal low pass filter to correctly work, while a linear interpolator, by construction has $h[n]$ that is non-ideal.

Suppose to have a sinusoidal input signal:

$$x[nT] = X_M \cos(2\pi f_0 nT)$$

And an interpolation filter system $L = 3$
The DTFT is ($\omega_0 = 2\pi f_0 T$)

$$X(e^{jw}) = \frac{X_M}{2} \cdot 2\pi \left[\sum_{k=-\infty}^{\infty} \delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)\right]$$

Due to the symmetry of the DTFT we just look at $k = \{0, \pm 1\}$ since after that we reach $[-\pi, \pi]$ symmetry
Notice that

$$\delta(3\omega \pm \omega_0 - 2\pi k) = \frac{1}{3}\delta(\omega \pm \frac{\omega_0 + 2\pi k}{3})$$

Recall that by sampling the z-Transform we obtain: $W(e^{j\omega}) = X(e^{jwL})$ and therefore. From here we get

$$W(e^{j\omega}) = X(e^{-j3\omega}) = \frac{\pi X_M}{3}\left[\sum_{k=-1}^{1} \delta(\omega - \frac{\omega_0 + 2\pi k}{3}) + \delta(\omega + \frac{\omega_0 + 2\pi k}{3})\right]$$
$$\text{IDFT} \downarrow$$
$$w[nT_0] = X_M \cos(2\pi f_0 nT_0) + X_M \cos(2\pi(F_s - f_0)nT_0) + X_M \cos(2\pi(F_s + f_0)nT_0)$$

Here we clearly see the spectral images appearing. An ideal filter with cutoff frequency f_0 will remove those, however a non ideal one will just attenuate these (and add a phase).
The system is LTI but then why do we get these images? This is valid only for homogeneous systems, that is, same input and output domains. However this system isn't homogeneous.

We can define the **normalized transition bandwidth** as the difference of pass-band and stop band frequency limits over the cut-off frequency:

$$\alpha = \frac{f_s - f_p}{F_S/2}$$

And the error becomes

$$|e[nT']| = |y[nT'] - y_i[nT']| \leq \delta_p X_M + (L-1)\delta_s X_M$$

A bigger L value means more errors as there are more undesired frequencies.

The power of the sinusoidal (same as ideal interpolation) is $P_x = \frac{X_M^2}{2}$
And by superimposition the power with the error has following bound ($\delta_s = \delta_p = \delta$): $P_e \leq L\frac{\delta^2 X_M^2}{2}$
With the SNR: $SNR_{dB} = 10\log_{10}\frac{P_x}{P_e} \geq -10\log_{10}(L\delta^2)$

For the minmax procedure we obtain that

$$N \approx \frac{-10\log_{10}(\delta_s\delta_p) - 13}{7.3\alpha} = L\frac{SNR_{dB} + 10\log_{10} L - 13}{7.3\alpha}$$

Multistage realization

We can factor L and do one interpolation step for each factor. Here the discovered formulas work the same way.

Each following step will have further apart replicas since the spectrum is replicated every $F_S/\prod L_i$ and therefore $\alpha_{k>1} >> \alpha_1$. and thus $N_{k>1} << N_1$. In the case of 2 steps we end up with

$$N_1 + N_2 \approx \frac{1}{L_2}N$$

A multistage interpolator is more efficient, because it avoids designing one long, high-rate FIR filter and instead splits the work into cheaper filters at lower rates.

7.2) Decimation

Decimation is the opposite of interpolation: it keeps one sample every $M \in \mathbb{N}^+$.

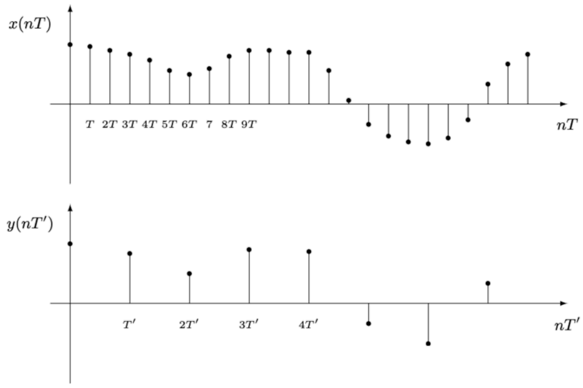
$$F'_S = \frac{1}{M}F_S \qquad T' = MT \qquad M \in \mathbb{N}^+$$

And the input/output equation is much simpler, since

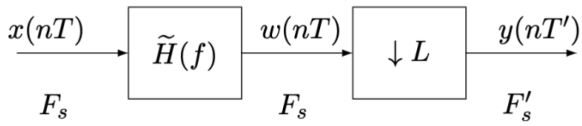
$$y[nT'] = x[nT'] = x[nMT]$$

Which is NOT time-invariant

Then, before decimating, we will add a **LPF with cutoff frequency $\mathbf{F'_s/2}$**



Example M=3



Complete System

To find the z-Transform we can define the signal $u[nT]$ as the result of an interpolation on $y[nT']$ with $L = M$ and normalized on $1/M$ and thus

$$u[nT] = \frac{1}{M} \cdot y[nT']_{\uparrow M} = \begin{cases} x[nT] & n = kM \\ 0 & x \neq kM \end{cases}$$

As seen before now we obtain

$$U(z) = \frac{1}{M}Y(z^M) \to Y(z) = MU(z^{1/M})$$

In terms of $X(z)$ we can see

$$u[nT] = x[nT]p[nT]$$

where

$$p[nT] = \begin{cases} 1 & n = kM \\ 0 & x \neq kM \end{cases} \qquad = \frac{1}{M}\sum_{k=0}^{M-1}(e^{-j\frac{2\pi}{M}})^{-kn}$$

Then the z-transform becomes (call $W_M = e^{-j\frac{2\pi}{M}}$):

$$U(z) = T\sum_{n=-\infty}^{\infty} x[nT]\cdot\frac{1}{M}\sum_{k=0}^{M} W_M^{-kn} z^{-n} = \frac{T}{M}\sum_{k=0}^{M-1}\sum_{n=-\infty}^{\infty} x[nT](zW_M^k)^{-n} = \frac{1}{M}\sum_{k=0}^{M-1} X(zW_M^k)$$

Where the last step was done by noticing that $T\sum_{n=-\infty}^{\infty} x[nT]zW_M^{k\ -n} = X(zW_M^k)$
So finally w have

$$Y(z) = \sum_{k=0}^{M-1} X(z^{\frac{1}{M}}W_M^k)$$

This is useful for computing the DFT as an the unit circle$z^{\frac{1}{M}} = e^{j2\pi fT'\frac{1}{M}} = e^{j2\pi fT}$

$$\tilde{Y}(f) = \sum_{k=0}^{M-1}\tilde{X}(f - kF_s')$$

Notice how $\tilde{Y}(f)$ is periodic with period $F_s' = \frac{1}{M}F_s$ and $\tilde{X}(f)$ with F_s. Therefore the decimated DFT must contain the same samples in less space. Now it is clear how the shift $-kF_s'$ behaves: it shifts the samples and some will superpose. Therefore $\tilde{X}(f)$ is not obtainable from $\tilde{Y}(f)$.

The only recovery is possible if $x[nT]$ is band limited in $[-\frac{F_s'}{M}, \frac{F_s'}{M}]$ as in this case no there si still enough "space" in the band of $\tilde{Y}(f)$ to fit them without superposing samples.

If we first use a low pass filter to artificially reduce the bandwidth to minimize the distortion. Now we have

$$Y(z) = \sum_{k=0}^{M-1} X(z^{\frac{1}{M}}W_M^k)H(z^{\frac{1}{M}}W_M^k)$$

$$\tilde{Y}(f) = \sum_{k=0}^{M-1}\tilde{X}(f - kF_s')\tilde{H}(f - kF_s') = \tilde{X}(f)$$

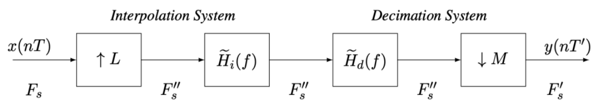
If the filter is non ideal, we get an error of form

$$\tilde{Y}(f) = \tilde{X}(f) + \tilde{Y}_a(f)$$

In this case the multistage realization yields the same advantages as before

7.3) Generic Sampling Rate Conversion

Let $L, M \in N^+$ be coprime. A Generic sampling conversion consists in changing the frequency by a rational constant $\frac{L}{M} \in \mathbb{Q}^+$



Complete System

Recall how

- The filter H_i is a low-pass filter with cutoff frequency $F_s/2$. That is used to reject the repetitions of $X(f)$ at frequencies kF_s with $k = 1, ..L-1$, in the interpolation stage.
- The filter H_d is a low-pass filter with cutoff frequency $F_s'/2$ that is used to avoid frequency aliasing in the decimation stage.

These two filters are cascaded and therefore it is possible to define a single low pass filter $H(f)$ that has cutoff frequency $f_p = \min\left\{\frac{F_s}{2}, \frac{F_s'}{2}\right\}$

$$\tilde{H}(f) = \begin{cases} 1 & |f| \leq \min\left\{\frac{F_s}{2}, \frac{F_s'}{2}\right\} \\ 0 & \min\left\{\frac{F_s}{2}, \frac{F_s'}{2}\right\} \leq |f| \leq \frac{F_s''}{2} \end{cases}$$

This system **is linear but not time-invariant**. It is LTI by considering period $T_{LTI} = LT' = MT = \text{least common multiple}(T, T')$

8) Infinite Impulse Response (IIR) Filters

As the name suggests these filters differ from FIR filters as their impulse response is not finite. This means that the output of the system uses past computed values for the computation of the current value.

$$y[n] = \sum_{l=1}^{N} a_l y[n-l] + \sum_{k=0}^{M} b_k x[n-k] = y_t[n] + y_{ss}[n]$$

So we need $N + M + 1$ coefficients to define the difference equation.

To solve such a system we need to also define the **two initial rest conditions**

1. The input must be considered zero before some starting time: $x[n] = 0 \; \forall n < n_0$
2. The output is likewise assumed to be zero before the signal starting time: $y[n = 0 \forall n < n_0$

The response to an unit impulse characterizes the LTI system completely: $y[n, x[n] = \delta[n]] = h[n]$

Therefore we can write the system in form

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Recall that due to feedback terms, an IIR used for a finite length system generally causes infinite length in output

Example.

Suppose to have a IIR system with first order difference equation:

$$y[n] = a_1 y[n-1] + b_0 x[n]$$

Then, by setting $x[n] = \delta[n]$ we end up with:

$$h[n] = a_1 h[n-1] + b_0 \delta[x]$$

Let the rest condition hold for $n < n_0 = 0$ then we can use the following table:

n	$n < 0$	0	1	2	3	4	...
$\delta[n]$	0	1	0	0	0	0	...
$h[n-1]$	0	0	b_0	$b_0(a_1)$	$b_0(a_1)^2$	$b_0(a_1)^3$	...
$h[n]$	0	b_0	$b_0(a_1)$	$b_0(a_1)^2$	$b_0(a_1)^3$	$b_0(a_1)^4$	...

Table

and finally

$$h[n] = \begin{cases} b_0(a_1)^n & n \geq 0 \\ 0 & n < 0 \end{cases} = b_0(a_1)^n u[n]$$

Example.

Let

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

then since the system is LTI we can use the previously obtained results to solve this system.

- *Linearity:* $y[n] = \mathcal{S}\{b_0 x[n] + b_1 x[n-1]\} = \mathcal{S}\{b_0 x[n]\} + \mathcal{S}\{b_1 x[n-1]\}$ *where the first result is known*

- *Time Invariance: this is used on the second term $y[n-1] = \mathcal{S}\{bx[n-1]\}$ which corresponds to $h[n-1]$*

Then we have that

$$h[n] = b_0(a_1)^n u[n] + b_1(a_1)^{n-1}u[n-1] = \begin{cases} 0 & n < 0 \\ b_0 & n = 0 \\ (b_0 + b_1 a^{-1})(a_1)^n & n \geq 1 \end{cases}$$

In the general case where the term $b_k x[n-k]$ is used we have

$$h[n] = b_k a_1^{n-1} u[n-k]$$

For finite length signals, by recalling the property of convolution with the dirac delta we have that

$$x[n] = (x * \delta)[n] = \sum_{k=n_0}^{N} x[k]\delta[n-k]$$

where $x[n]$ is a finite length signal with non zero samples in $[n_0, N]$.
then we have

$$y[n] = (x * h)[n] = (x * \delta * h)[n] = \big(x * (\delta * h)[n]\big)[n] = \sum_{k=n_0}^{N} x[k]h[n-k]$$

With an long (infinite) signal we can just iterate the system indefinitely.

Theorem 17.

FIR filters always have polynomial system function $X(z)$, but when there is feedback, also $Y(z)$ is a polynomial and thus

$$y[n] = (h * x)[n] \iff Y(z) = H(z)X(z) \to H(z) \text{ is a rato fo two polynomials}$$

Remark.

Convolution and difference equation are the same ONLY for FIR

In IIR filters where the convolution is often not practical

Example.

Recall

$$y[n] = ay[n-1] + b[x] \to h[n] = b(a)^n u[n]$$

Now set $x[n] = u[n]$. Here iterating the difference equation is equivalent to "multiplying the previous output by a and adding b" since $bx[n] = bu[n] = b\ n \geq 0$.

n	$x[n]$	$y[n]$
$n < 0$	0	0
0	1	b_0
1	1	$b_0 + b_0(a_1)$
2	1	$b_0 + b_0(a_1) + b_0\,(a_1)^2$
3	1	$b_0(1 + a_1 + a_1^2 + a_1^3)$
4	1	$b_0(1 + a_1 + a_1^2 + a_1^3 + a_1^4)$
$\vdots$	1	$\vdots$

Table

That is

$$y[n] = b\sum_{k=0}^{n} a^k = b\frac{1-a^{n+1}}{1-a}$$

Where 3 cases might happen:

- $|a| > 1$*: the numerator dominates and the output keeps growing. This is* ***unstable*** *and is a situation to avoid*
- $|a| < 1$*: in this case the system is stable as* a^{n+1} *tends to zero and the utput has a limiting value* $y[n \to \infty] = b/1-a$
- $|a| = 1$*: is generally unbounded but not always*

Now let's try the convolution sum

$$\begin{aligned} y[n] = (u * h)[n] &= \sum_{k=-\infty}^{\infty} b(a)^{n-k}u[n-k]u[n] = \sum_{k=0}^{n} b(a)^{n-k} \\ &= ba^n\sum_{k=0}^{n} a^{-k} = ba^n\sum_{k=0}^{n}(\frac{1}{a})^k \\ &= ba^n\frac{1-\frac{1}{a}^{n+1}}{1-\frac{1}{a}} = b\frac{1-a^{n+1}}{1-a} \quad n \geq 0 \end{aligned}$$

8.1) General First-Order Case

Let this be the general form of the first-order difference equation with feedback:

$$y[n] = a_1 y[n-1] + b_0 x[n] + b_1 x[n-1]$$

Then the z-Transform is:

$$Y(z) = a_1 z^{-1} Y(z) + b_0 X(z) + b_1 z^{-1} X(z)$$

And thus the transfer function is

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1}}{1 - a_1 z^{-1}}$$

Remark (Property of Time Shifted Z-Transform)**.**

$$\text{Time Delay: } n_0 > 0 \quad x'[n] = x[n-n_0] \xrightarrow{z} X'(z) = z^{-n_0}X(z), \quad R_{x'} = R_x$$

Theorem 18.

*The coefficients of the **numerator** polynomial of the system function of an IIR system are the **coefficients of the feed-forward** $x[n]$ **terms** of the difference equation.*

*For the **denominator** polynomial, the constant term is one, and the remaining coefficients are the **negatives of the feedback coefficients** $y[n-k]$.*

Also recall

$$x[n] = a^n u[n] \leftrightarrow X(z) = \frac{1}{1 - az^{-1}}, \quad R_x = \{|a| < |z|\}$$

8.2) IIR Filter Design (Transformation $s \to z$ Method)

IIR filters can be designed in various ways. we will focus on the **Transformation $s \to z$ method on magnitude only specifications.**

This method consists in first finding the **desired frequency response** $D(e^{j\omega})$, then use a mapping $s \to z$ to get the **equivalent analog desired response** $D_a(j\omega)$. Using well known analog design techniques we find $H_a(j\omega) \approx D_a(j\omega)$ and through the inverse mapping $z \to s$ we obtain $H(e^{j\omega}) \approx D(e^{j\omega})$.

This method is effective as analog systems are all of type $H(s) = \frac{Z(s)}{P(s)}$ which correctly translate to IIR filters. Moreover analog design has better analytical solutions.

Definition 19 (Bilinear Transformation)**.**

First let's define the properties that the transformation function must have:

- rational transfer functions in s must be mapped to rational transfer functions in z
- stable tf in s must be mapped to stable tf in z
- the imaginary axis must be mapped to the unit circle

It turns out that the **Möbius Transformation** satisfies these properties

$$s = \frac{\alpha z + \beta}{\gamma z + \delta}$$

with $\alpha, \beta, \gamma \in \mathbb{C}$ and $\alpha\delta - \beta\gamma \neq 0$

Generally we set the **bilinear transformation** as:

$$s = \frac{z-1}{z+1} \leftrightarrow z = \frac{1+s}{1-s}$$

Moreover we have that

- $s = 0 \leftrightarrow z = 1$: that is, zero frequency in s is same as zero frequency in z
- $s = \infty \leftrightarrow z = -1$: that is the max frequency in s in the max in z
- $s = -1 \leftrightarrow z = 0$: that is BIBO stability is preserved $\Re[s] < 0 \leftrightarrow |z| < 1$

So, once we have the analog solution, we must use this transformation to approximate $H(z)$:

$$H(e^{j\omega}) = H(z) = H_a\left(\frac{z-1}{z+1}\right) = H_a(s \to z)$$

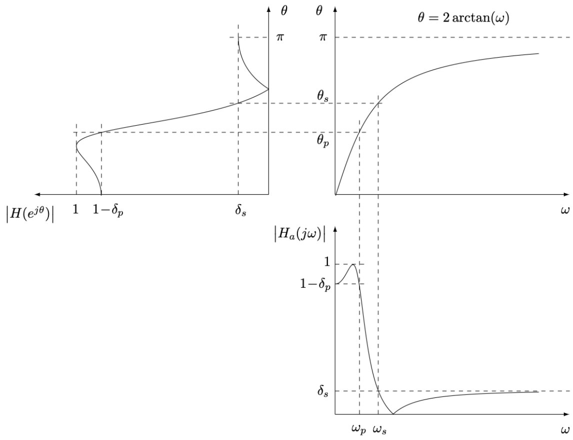
The mapping clearly satisfies the unit circle↔imaginary axis property, moreover it is necessarily non-linear as it has to map infinite values on the imaginary axis into finite values on the unit circle. There is a distortion on the frequency axis (but not magnitude). Finally our filter acquires the following specifications:

Pass-Band:

$$\begin{array}{l} 0 \le \omega \le \omega_p = \tan\theta_p/2 \\ 1-\delta_p \le |H_a(j\omega)| \le 1 \end{array} \implies \begin{array}{l} 0 \le \theta \le \theta_p = 2\arctan\omega_p \\ 1-\delta_p \le |H_a(e^{j\theta})| \le 1 \end{array}$$

Stop-Band:

$$\begin{array}{l} \omega \ge \omega_s = \tan\theta_p/2 \\ |H_a(j\omega)| \le \delta_s \end{array} \implies \begin{array}{l} \theta_s = 2\arctan\omega_s \le \theta \le \pi \\ |H_a(e^{j\theta})| \le \delta_s \end{array}$$



Plots

Quick proof:
Let's see how the mapping behaves starting from the unit circle $z = e^{j\theta}$:

$$s \overset{z=e^{j\theta}}{=} \frac{e^{j\theta}-1}{e^{j\theta}+1} = \frac{e^{j\frac{\theta}{2}}(e^{j\frac{\theta}{2}} - e^{-j\frac{\theta}{2}})}{e^{j\frac{\theta}{2}}(e^{j\frac{\theta}{2}} + e^{-j\frac{\theta}{2}})} = \frac{2j\sin(\theta/2)}{2\cos(\theta/2)} = j\tan(\theta/2) = j\omega$$

with $\omega = \tan\theta/2$ and $\theta = 2\arctan\omega$. This shows the non-linearity in the mapping.

Also vice versa (to show circle→axis):

$$z = \frac{j\omega+1}{j\omega-1} = |z|e^{j\varphi(z)} \to \begin{cases} |z| = \frac{|j\omega+1|}{|j\omega-1|} = \frac{\sqrt{1+\omega^2}}{\sqrt{1+\omega^2}} = 1 \\ \varphi(z) = \angle(1+j\omega) - \angle(-1+j\omega) = \arctan(\omega) - (\pi - \arctan(\omega)) \end{cases}$$
$$\to z = e^{j(2\arctan(\omega)-\pi)}$$

The mapping is satisfied but with an initial phase shift added.

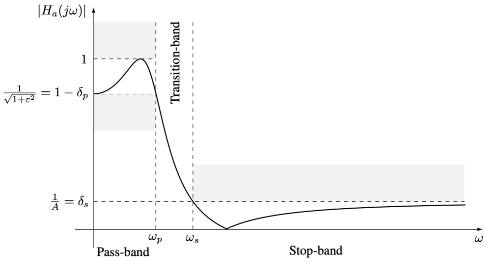
□

> **Remark.**
>
> The magnitude is preserved also in behaviour.
> The phase is not linear, except for very small values where $\tan(\theta/s) \approx \theta/2$.

8.3) Analog Filter Solutions

We will design a *passive* analog low pass filter in which $\max_\omega |H_a(j\omega)| = 1$. From here we can redefine

$$1 - \delta_p = \frac{1}{\sqrt{1+\epsilon^2}} \qquad \rightarrow PBR_{dB} = 20\log_{10}\sqrt{1+\epsilon^2}$$
$$\delta_s = \frac{1}{A} \qquad \rightarrow A_{dB} = 20\log_{10} A$$



Example

Moreover let's define two classes of filters:
All-Pole Filters:

$$H_a(s) = \frac{1}{D_a(s)}$$

where $D_a(s)$ is a polynomial in s. These filters ensure that

$$\lim_{\omega \to \infty} |H_a(j\omega)| = 0$$

All zeroes are at infinity. These are used for Butterworth and Chebyshev Type 1 filters.

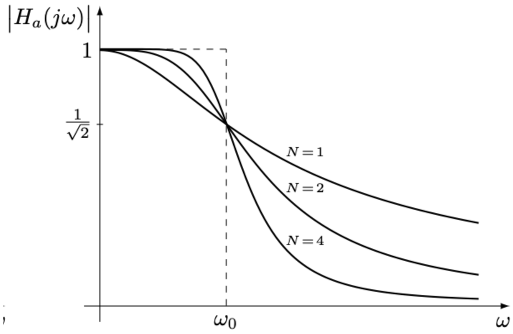
Poles and Zeros Filters:

$$H_a(j\omega) = \frac{N_a(j\omega)}{D_a(j\omega)}$$

Where $\deg N_a(s) \leq \deg D_a(s)$. And there can be finite zeros. These are used for Elliptic (Cauer) and Chebyshev Type 2 filters.

Butterworth Filters

Butterworth have **maximally flat magnitude at zero frequency**:



Example

It has following squared magnitude function:

$$M_a(j\omega) = |H_a(j\omega)|^2 \frac{1}{1 + \left(\frac{\omega}{\omega_0}\right)^{2N}}$$

Where ω_0 is the 3dB cutoff frequency.
Notice how as $N \to \infty$ we have that $|H_a(j\omega)| \to u(\omega - \omega_0)$

There are $2N$ poles on the circle of radius ω_0 which are given by $s_k = \omega_0 e^{j\pi\left(\frac{1}{2} + \frac{2k-1}{2N}\right)}$

We can estimate the order N from the specifications:

$$\left(\frac{\omega_p}{\omega_0}\right)^{2N} = \epsilon^2 \qquad \rightarrow N = \frac{1}{2}\frac{\log_{10}\frac{A^2-1}{\epsilon^2}}{\log_{10}\omega_s/\omega_p}$$
$$\left(\frac{\omega_s}{\omega_0}\right)^{2N} = A^2 - 1$$

> **Remark.**
>
> A Butterworth filter is completely specified by N and ω_0.

It is always monotone.

Here is a quick proof:
The squared magnitude off $D_a(j\omega)$ is a polynomial of degree N in ω^2 since

$$|D_a(j\omega)|^2 = D_a(j\omega)D_a^*(j\omega) = D_a(j\omega)D_a(-j\omega)$$

and clearly all odd powers cancel out.
Let's call it $P_N(\omega^2) = p_0 + p_1\omega^2 + \ldots + p_N\omega^{2N}$. The maximally flat constraint forces to have $p_{0\div N-1} = 0$ and thus $P_N(\omega^2) = 1 + p_N\omega^{2N}$

Then if we find the magnitude squared we have

$$M_a(j\omega) = |H_a(j\omega)|^2 = \frac{1}{P_N(\omega^2)}$$

where $P_N(\omega^2)$ is a polynomial of degree N in ω^2. Since the filter is real, the magnitude squared is an even function.

Notice that by design we want that $|H_a(0)| = 1$ as the maximum and also for the flatness we set $P_N(\omega) = 1 + \omega^{2N}$
Since ω_0 is the 3dB cutoff frequency we have that

$$|H_a(\omega)| = 3[dB] \approx \frac{1}{2}[\text{magnitude sq}] = \frac{1}{1+\omega^{2N}} \rightarrow \omega = 1$$

And since it should work for all ω_0 we set

$$M_a(j\omega) = |H_a(j\omega)|^2 \frac{1}{1 + \left(\frac{\omega}{\omega_0}\right)^{2N}}$$

Here the poles are also explicited since by looking at the denominator:

$$1 + \frac{s}{j\omega_0}^{2N} \rightarrow \frac{s}{j\omega_0}^{2N} = -1 \rightarrow s^{2N} = (\omega_0 e^{j\frac{\pi}{2}})^{2N} e^{j(\pi+2k\pi)} \rightarrow s_k = \omega_0 e^{j(\frac{\pi}{2}+\frac{\pi+2k\pi}{2N})}$$

By looking at the exponent it is clear that

$$\frac{\pi}{2} + \frac{\pi+2k\pi}{2N} = \pi(\frac{1}{2} + \frac{2k+1}{2N})\ \forall k = 0,..,2N-1$$

to get the formula I wrote call $k' = k+1$

□

Chebyshev Type 1

This filter is obtained by a min/maxing method and results in the following magnitude squared:

$$M_a(j\omega) = \frac{1}{1+\epsilon^2 \cdot T_N^2(\omega/\omega_p)} \text{ where } T_N = \begin{cases} \cos(N \arccos(x)) & |x| \leq 1 \\ \cosh(N \text{arccosh}(x)) & |x| > 1 \end{cases}$$

And T_N can be recursively rewritten as

$$T_{N+1}(x) = 2xT_N(x) - T_{N-1}(x) \qquad T_0(x) = 1, T_1(x) = x$$

This polynomial has the following properties:

- T_N is even when N is even, otherwise it is odd and $T_N(1) = 1$ and then $T_N(-1) = \pm 1$
- It has N distinct and real zeroes in the interval $(-1, 1)$ and takes value ± 1(only one) between two consecutive zeroes
- It is the solution to the minmax problem: $\min_{P_N} \max_{|x|\leq 1} |P_N(x)|$

From the last property it is clear that this filter best approximates a low-pass filter. The poles are on an ellipse on the s plane

It has an equiripple in the passband but monotone in stopband.

Quick proof:
Set $x = \cos(\varphi) < 1 \; \forall \varphi$ and then $T_n(\cos(\varphi)) = cos(N\varphi)$. Now set $\varphi = \arccos x$ and obtain
TODO

□

Chebyshev Type 2

It has is monotonic in the passband but has equiripple in stopband.
We reuse the prior studied polynomial:

$$M_a(j\omega) = \frac{1}{1+\epsilon^2 \dfrac{T_N^2(\omega_s/\omega_p)}{T_N^2(\omega_s/\omega)}}$$

9) A Discussion About Stability, Poles and Z-Transform

In this course we mainly discussed causal systems. The rules about stability can be recapped as

Suppose $y[n] = y_t[n] + y_{ss}[n]$ and the system be causal where the behaviour of the two responses is known, then:

- If ALL the poles are inside the unit circle $|p_i| < 1 \; \forall i$ we have that $y_t[n] \to 0$. Here a bounded input will result in a bounded output (BIBO)
- If any pole is on the unit circle $|p_i| = 1$ we have $y_t[n]$ oscillatory (or constant) and the the system is *marginally* stable since it diverges if the input resonates with a pole, otherwise it stays bounded
- If any pole is outside the unit circle $|p_i| > 1$ we have an exponential growth in the transient response, this is not BIBO
- If there is a multiplicity of poles on the unit circle it also is never stable (the poles resonate with eachother)

A quick discussion on $|p_i| = 1 \; \forall i$:
Suppose there are n non overlapping poles, then $Y_t(\omega_k) \neq 0 \quad \forall k = \{0, \dots, n-1\}$ where ω_k are the n frequencies of the poles. Let $X(\omega_k) = 0$ then the input doesn't resonate with the poles and the output will be bounded. However if $X(\omega) \neq 0$ for at least one value of ω_k then it will diverge

However the z-Transform gives us the main rule on BIBO stability independently on the causality of the system, that is:

$$\boxed{\text{BIBO stability} \iff \text{ROC includes unit circle}}$$

Proof:
For a causal system this is clearly true since

$$|p_i| < 1 \iff \text{ROC includes unit circle}$$

since ROC is $|z| > \max_i\{|p_i|\}$

For an anti-causal system we have ROC $|z| < \min_i\{|p_i|\}$ and thus we can "reverse" the rules of the causal system.

For a two-sided system the ROC will be the intersections between the causal ROC and the anticausal part. Clearly the ROC exists only if $\min_i\{\text{anti-causal } |p_i|\} > \min_j\{\text{causal } |p_j|\}$ otherwise there will be no intersection, and thus if the poles follow the rule for the causal and anticausal part the ROC will include $|z| = 1$.

□

Now here is the tricky part: **an inverse z-Transform is uniquely defined by $\mathbf{H(z)}$ and the ROC**, this means that $H(z)$ alone doesn't specify $h[n]$.

Example.

Why do we need the ROC? Suppose I have $H(z) = a\frac{1}{1-az^-1}$ *this is equivalent to* $H(z) = -\frac{az}{a-z}$

- *The first transform has inverse* $h[n] = a(a)^n u[n] = (a)^{n+1}u[n]$ *and is causal with ROC* $|z| > |a|$
- *The second transform (which is equivalent) has inverse* $h[n] = -(a)^{n+1}u[-n-1]$ *which is anti-causal with ROC* $|z| < |a|$

So it depends on what you aim for:
If you want a causal system you choose the first inverse (which can e unstable)
If you want a stable system you choose the correct inverse that contains the unit circle (this might be anticausal)